

22050 Continuous-Time Signals and Linear Systems

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L07

ADC performance

Sampling

Aliasing

Amplitude quantization

Quantization noise

Oversampling and averaging

Matching anti-aliasing filter to ADC

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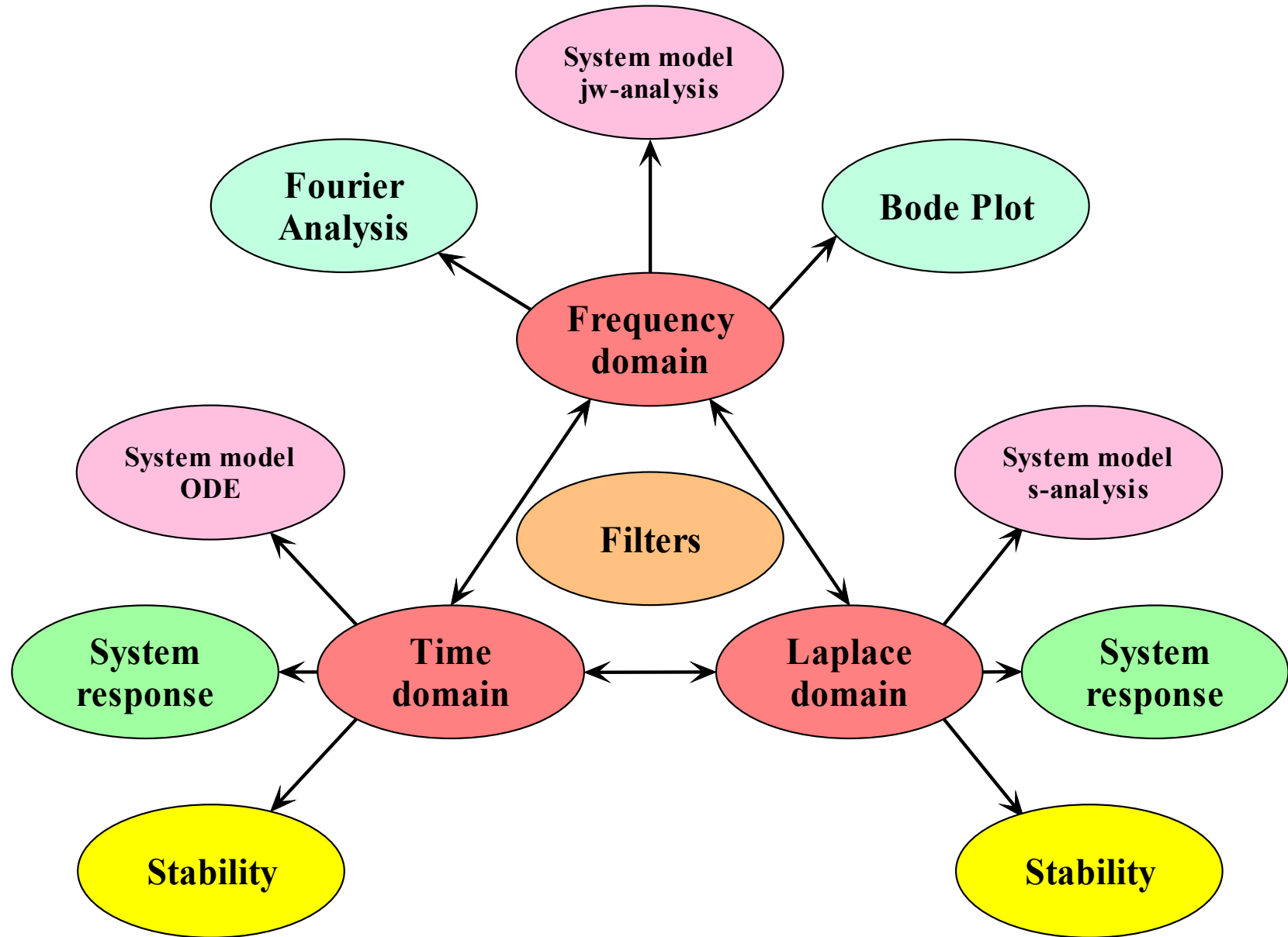
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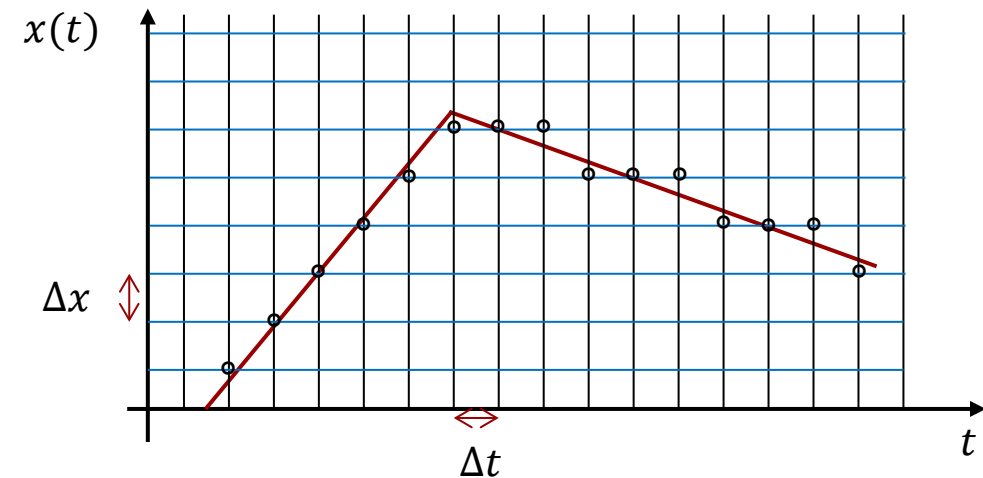
- Analogue-digital conversion
 - Sampling
 - Aliasing
 - Amplitude quantization
 - Truncating ADC
 - Rounding ADC
 - Quantization noise
 - RMS value of signal with non-zero mean
 - Effects of averaging on mean and variance of a random variable
 - RMS value of quantization noise in truncating ADC
 - RMS value of quantization noise in rounding ADC
 - Signal to noise ratio
 - Truncating ADC
 - Rounding ADC
 - Effect of oversampling and averaging
 - Inadequate amplification
 - Effective number of bits
 - Matching anti-aliasing filter to ADC
 - Determining design specifications for anti-aliasing filter

Analogue-Digital Conversion

Aliasing

Video

Signals originating from physical processes are typically continuous in both time and amplitude. Such signals are said to be analogue. Examples of analogue signals are the ambient temperature, blood pressure and sound pressure from speech. An example of a signal that is not continuous in time is the heart rate interval. It only has a new value after each heartbeat, but no value in between heart beats. Such a signal is a discrete-time (D-T) signal.



The red curve in the graph is an analogue signal. We can make sample recordings at equidistant time intervals Δt marked by the thin vertical lines. This process is called **SAMPLING**.

If we are to store the amplitude of each sample using binary numbers, we only have a finite number of amplitude values, that we can store, here marked by the thin horizontal lines. Finding that binary value representable in the microprocessor which is closest to the amplitude of the analogue signal is called **QUANTIZATION**. The black circles indicate the result of **rounding/truncating** the amplitude of each sample to its nearest binary value. The vertical distance between the circle (binary value) and the red curve (analogue value) is called **quantization error**.

We will start the discussion of Analogue-Digital conversion by investigating the significance of Δt and Δx .

Quantization:

Signal amplitudes are rounded **or truncated** to nearest possible number

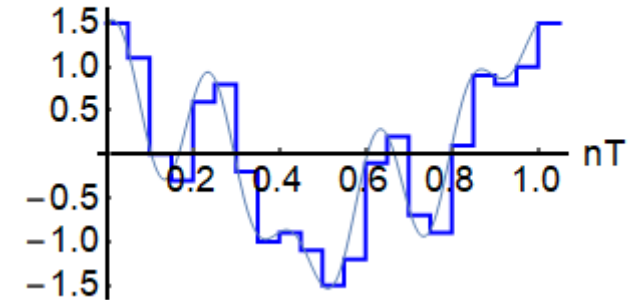
Sampling:

Signal amplitudes are sensed for a brief moment and the value held in a **Sample-&Hold** circuit until conversion is complete.

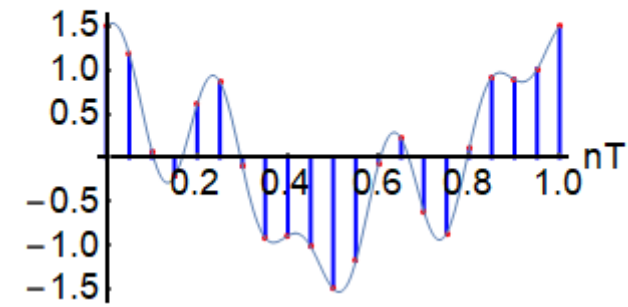
Sampling and quantization

The ADC **rounds or truncates** the amplitude of the time-discrete samples to nearest possible number.

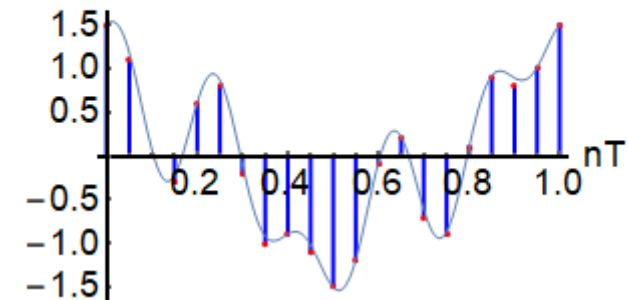
Continuous time/Discrete amplitude



Discrete time/Continuous amplitude



Discrete time/Discrete amplitude



Sampling saves discrete values of a signal:

$$x(t) \rightarrow x(n \cdot \Delta t)$$

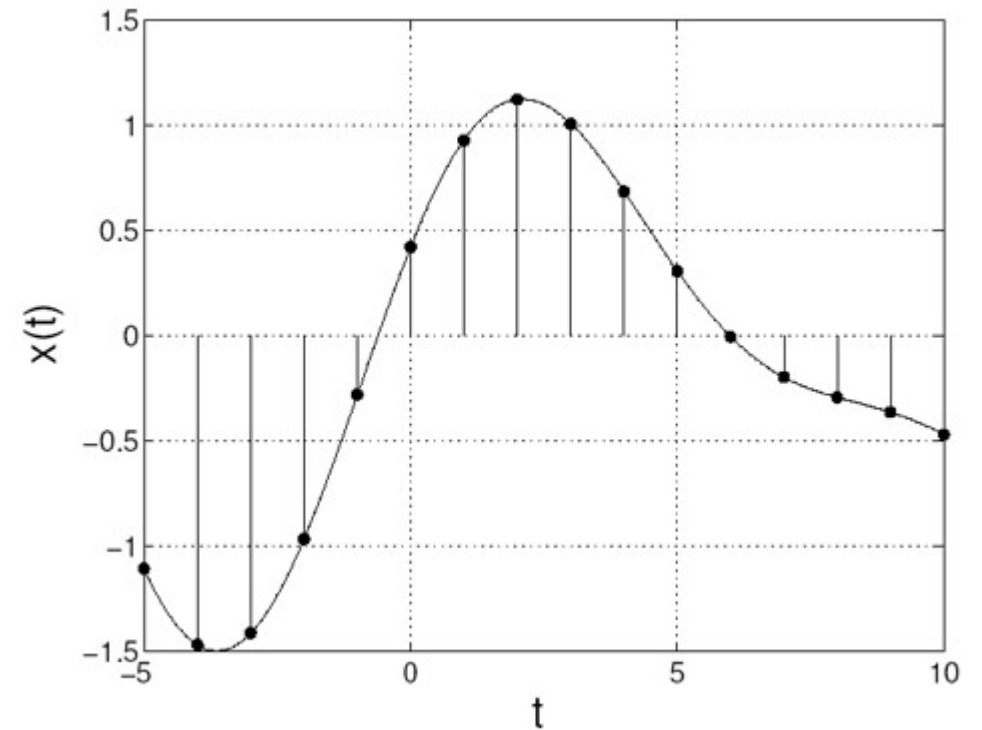
The time interval between sampling is the sampling interval, and its reciprocal is the **sampling frequency**:

$$F_S = \frac{1}{\Delta t}$$

Nyquist sampling theorem:

A unique reconstruction of a sinusoidal signal from its samples requires more than two samples in each period:

$$\Delta t < T/2 \Leftrightarrow F_S > 2f_{\max}$$



We know that the cos function is an even function:

$$x(t) = A \cos(2\pi f \cdot t) = A \cos(2\pi |f| \cdot t)$$

We make a sampled copy x_k of the signal:

$$x_k(n\Delta t) = A \cos(2\pi \cdot |f_k| \cdot n\Delta t)$$

$$\Delta t = \frac{1}{f_s}$$

$$x_k(n\Delta t) = A \cos\left(2\pi \cdot |f_k| \cdot \frac{n}{f_s}\right)$$

$$x_k(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \frac{|f_k|}{f_s}\right)$$

We can add a multiple of 2π :

$$x_k(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \left(\frac{|f_k|}{f_s} + m \frac{f_s}{f_s}\right)\right)$$

We observe that if we add a multiple of the sampling frequency f_s to the signal frequency f_k , we will get the exact same sample values.

$$x_k(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \left(\frac{|f_k| + m f_s}{f_s}\right)\right)$$

The signal frequency f_k is said to have **aliases** in $f_k + m \cdot f_s$

An example:

A 1 Hz cosine sampled at 6 Hz:

$$x_1(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \frac{1}{6}\right) = A \cos\left(2\pi \cdot n \cdot \frac{-1}{6}\right)$$

A 5 Hz cosine sampled at 6 Hz:

$$x_5(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \frac{5}{6}\right) = A \cos\left(2\pi \cdot n \cdot \left(\frac{-1+6}{6}\right)\right)$$

$$x_5(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \left(\frac{-1}{6}\right) + 2\pi n\right) = x_1(n\Delta t)$$

A 7 Hz cosine sampled at 6 Hz:

$$x_7(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \frac{7}{6}\right) = A \cos\left(2\pi \cdot n \cdot \left(\frac{1+6}{6}\right)\right)$$

$$x_7(n\Delta t) = A \cos\left(2\pi \cdot n \cdot \left(\frac{1}{6}\right) + 2\pi n\right) = x_1(n\Delta t)$$

Consequence of inadequate sampling frequency

Top: 1 Hz sinusoid sampled at a rate 6 Hz.

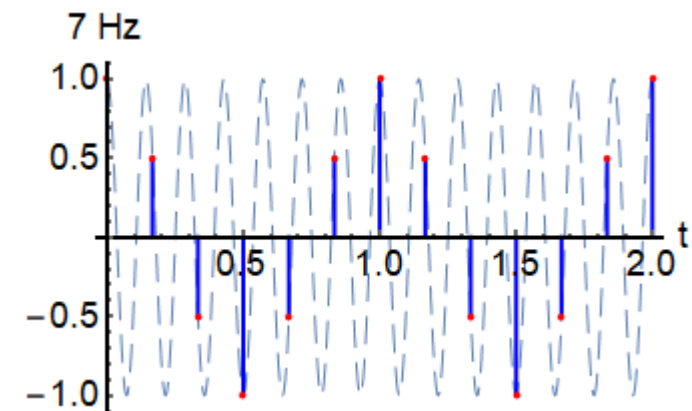
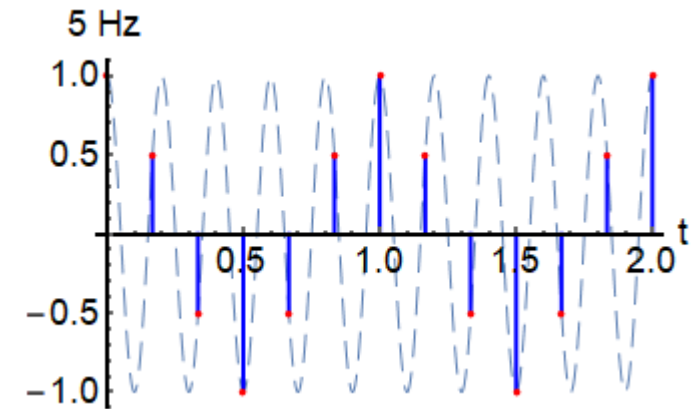
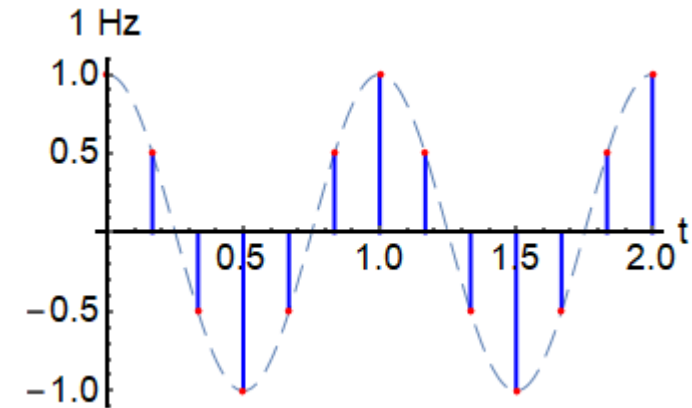
Center: 5 Hz sinusoid sampled at a rate of 6 Hz.

Bottom: 7 Hz sinusoid sampled at a rate of 6 Hz.

We see that even though we have sampled three different sinusoids, **the samples come out identical** in the three cases. If we were given the set of samples and were asked to draw the continuous time sinusoid through the samples, we would be faced with an ambiguous answer: should we draw a 1 Hz, a 5 Hz or a 7 Hz sinusoid?

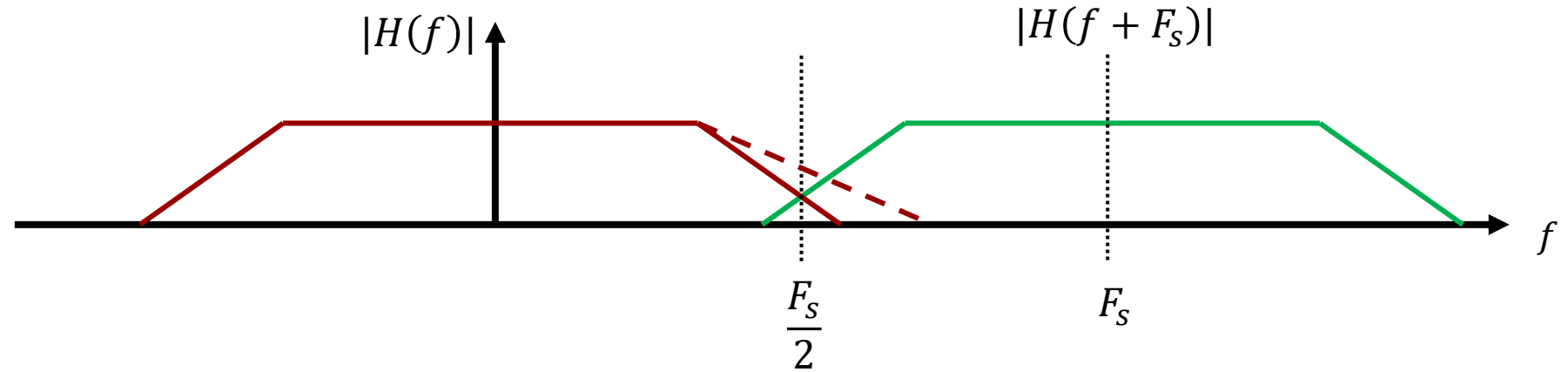
The three signals are said to be each other's alias and the issue is called **aliasing**.

If we assume that we have adhered to the Nyquist sampling theorem, we may conclude that the samples represent a 1 Hz sinusoid, not a 6 or 7 Hz sinusoid.

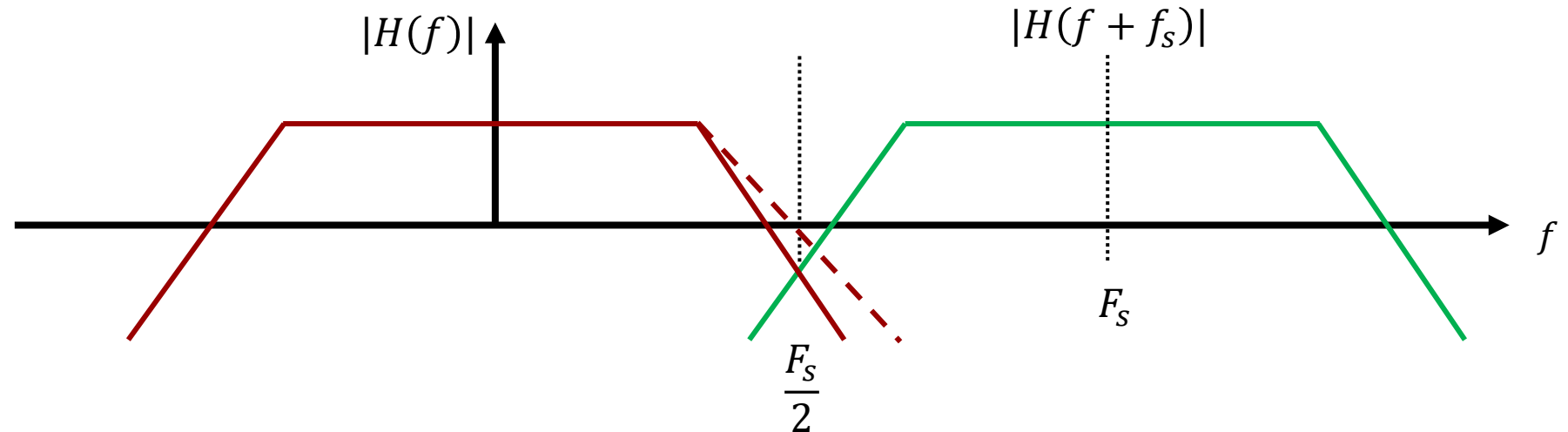


Reconstruction of Continuous-Time Signal

Aliasing
increases energy
around $F_s/2$.



Effect of aliasing
reduced by lowpass
filtering.

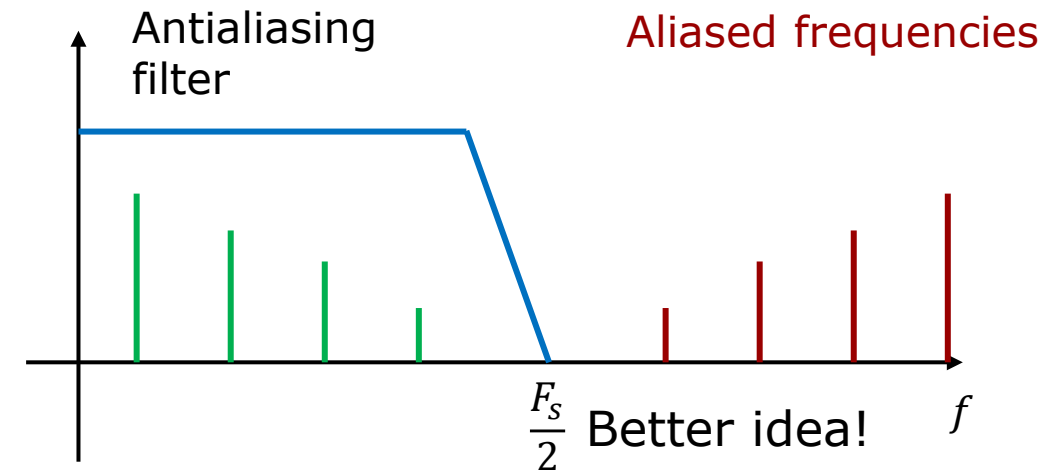
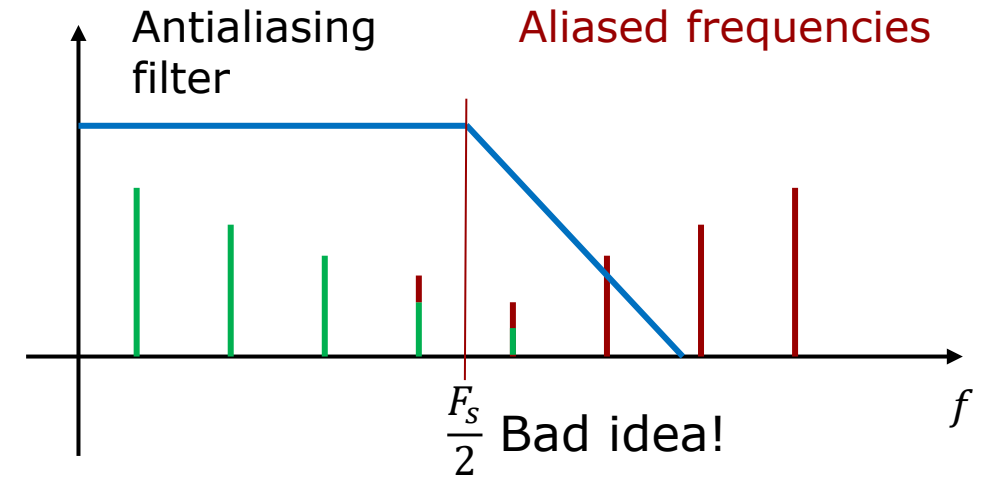


The aliased frequencies will fold down and add to the fundamental frequencies (**green stems**) making their amplitudes higher than they are without aliasing.

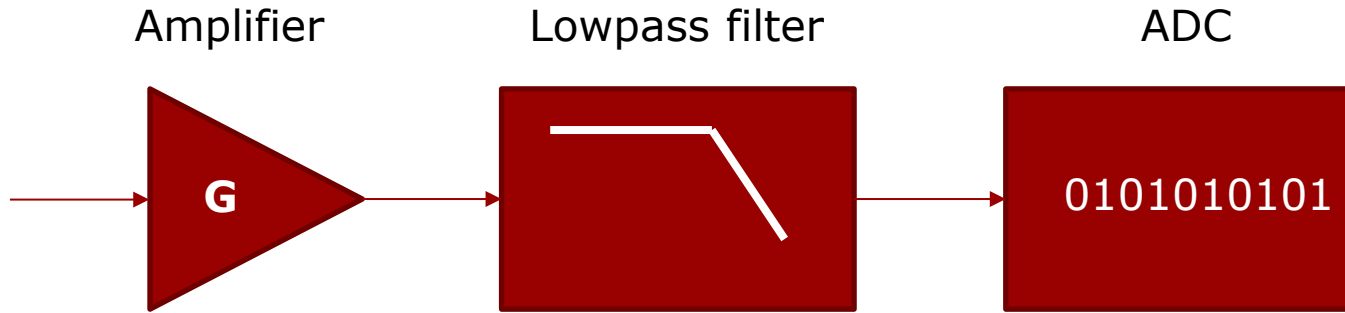
If we can dampen the magnitude of all frequencies above $F_s/2$, then we can reduce how much the aliased frequency add to the fundamental frequencies.

In the top figure we use a lowpass filter to dampen aliased frequencies, but the sampling frequency is too low compared to the cut-off frequency of the filter.

In the bottom figure, the sampling frequency has been increased, and the filter order has been increased as well.



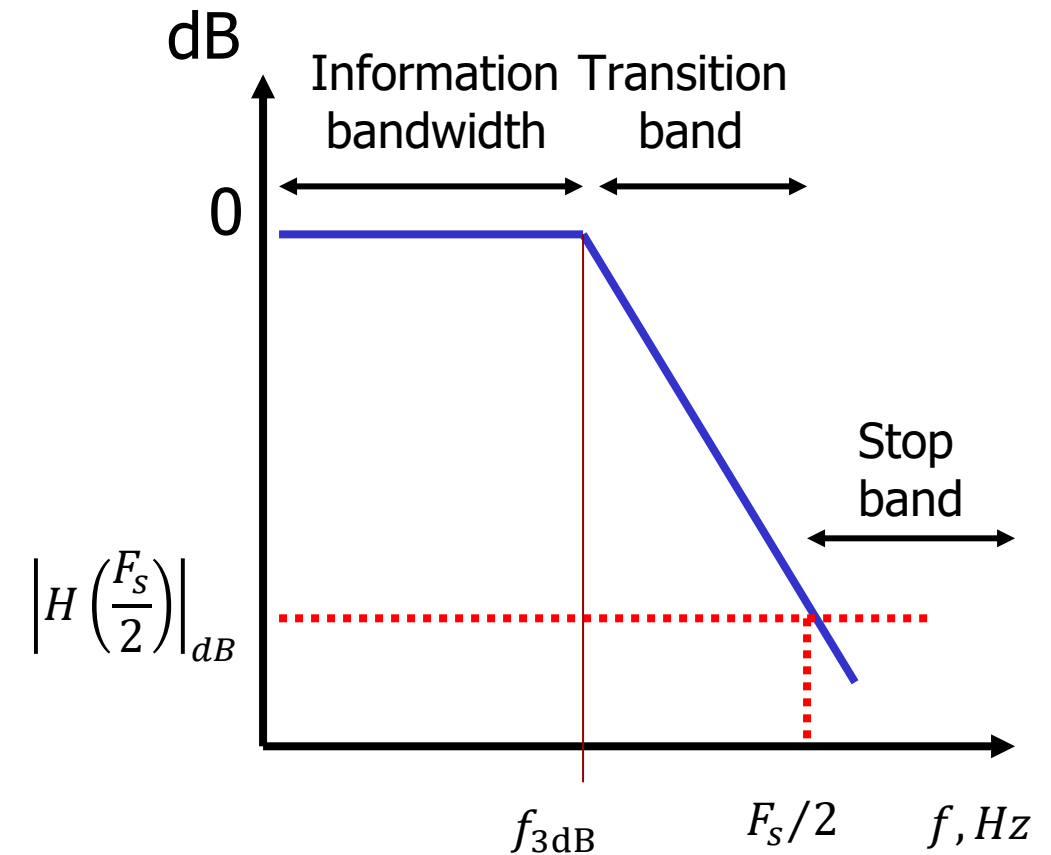
Anti-aliasing filter



Setting the sampling frequency to two times the 3dB cut-off frequency of the lowpass filter is not good enough because the signal is only attenuated about 3-6 dB at the cut-off frequency.

We could benefit from some guidance to setting an attenuation level for $F_s/2$. In other words, what level of damping should the anti-aliasing filter achieve at $F_s/2$?

The ADC has a finite number of bits. Hence there is a limit to how small signal amplitudes the ADC can “see”. We need to investigate, if the ADC can give some suggestions to how much the lowpass filter should attenuate the signal at $F_s/2$.



Analogue-Digital Conversion

Amplitude quantization

Video

To understand the conversion process, we will study an ADC with 4 bits.

The table on the right shows that with 4 bits we can represent $2^4 = 16$ values.

Here a **reference voltage of 5V** is used, hence the voltage range from 0V to 5V is divided into **16 intervals**.

The voltage range of one interval is called the **code width**.

We will denote this voltage interval $\Delta \stackrel{\text{def}}{=} V_{LSB}$. It is the voltage increase necessary to toggle the least significant bit (LSB).

$$\text{Code width: } \Delta \stackrel{\text{def}}{=} V_{LSB} = \frac{V_{range}}{2^{\#bit}} = \frac{5V}{2^4} = 0.3125V$$

We must remember, that a certain bit pattern represents a **voltage range**, not a specific voltage. Since we in this case have 16 bit-patterns, we also get 16 voltage intervals.

There are several different ADC principles for setting the bits. We will only consider the successive approximation register (SAR) ADC.

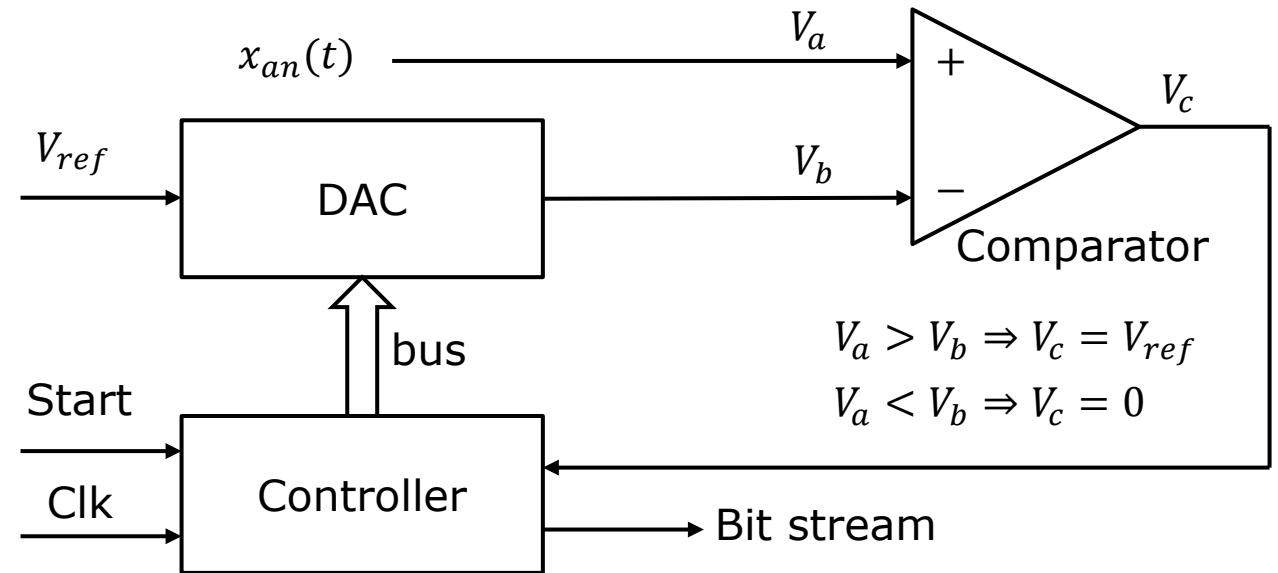
bit	3	2	1	0	Value
0	0	0	0	0	0,00000
1	0	0	0	1	0,31250
2	0	0	1	0	0,62500
3	0	0	1	1	0,93750
4	0	1	0	0	1,25000
5	0	1	0	1	1,56250
6	0	1	1	0	1,87500
7	0	1	1	1	2,18750
8	1	0	0	0	2,50000
9	1	0	0	1	2,81250
10	1	0	1	0	3,12500
11	1	0	1	1	3,43750
12	1	1	0	0	3,75000
13	1	1	0	1	4,06250
14	1	1	1	0	4,37500
15	1	1	1	1	4,68750
1	0	0	0	0	5,00000

Conversion principle of truncating SAR ADC

Successive approximation register (SAR) ADC.

Starting from the most significant bit down to the least significant bit, the controller turns on each bit one at a time and generates an analogue signal (V_b), with the help of the digital-to-analogue converter (DAC), to be compared in the comparator with the original input analogue signal (V_a). Based on the result of the comparison (V_c), the controller changes or leaves the current bit and turns on the next most significant bit. The process continues until decisions are made for all available bits. The result is a bit pattern corresponding to 2.8125V, which is 0.1875V lower than the analogue value.

Observation: The result is always lower than the analogue value. We will call this a **truncating ADC**.



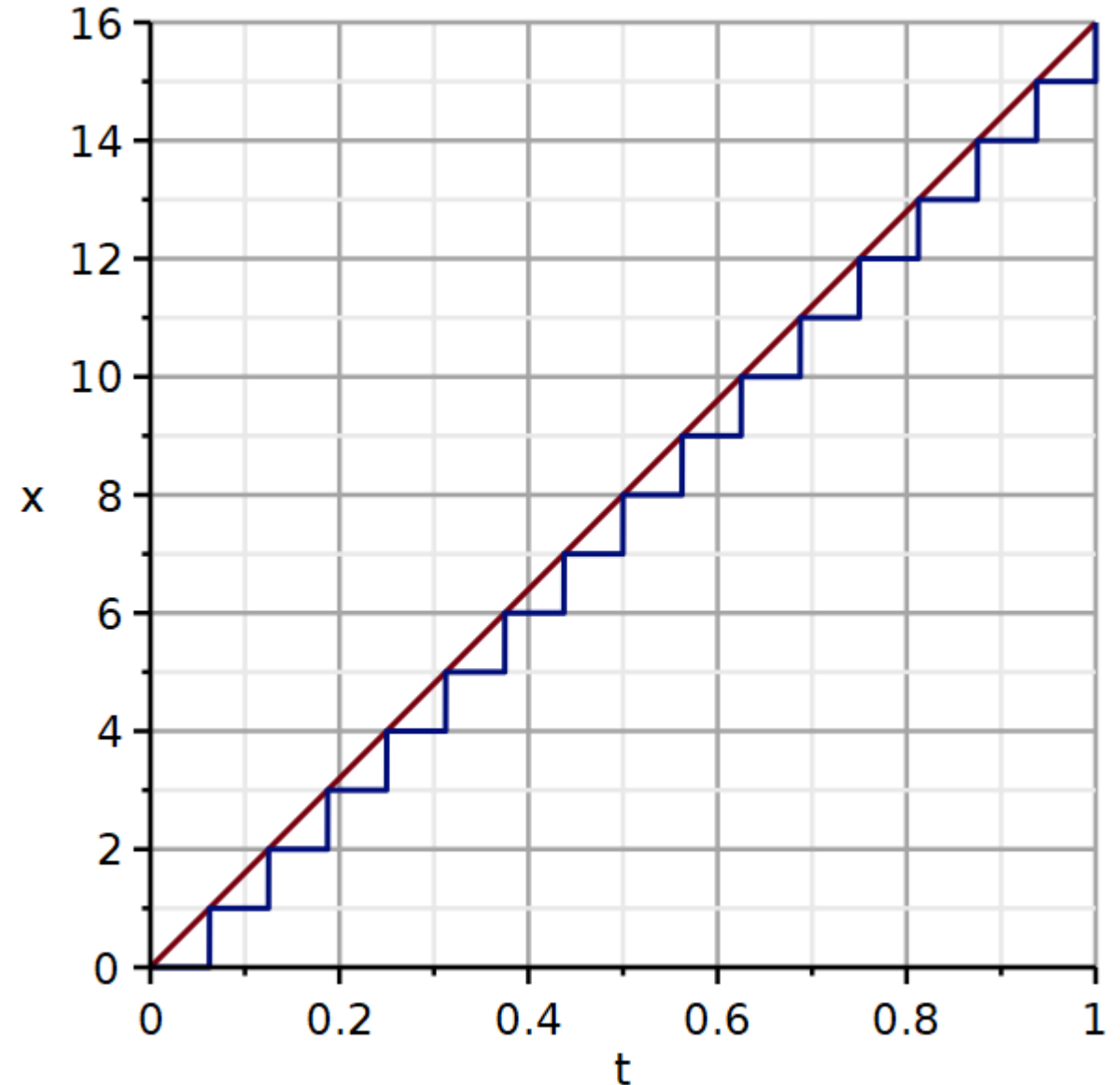
4-bit converter, $V_{ref} = 5V$

V_a (V)	V_b (V)	V_c (V)	MSB				LSB
			3	2	1	0	
3	2,500	5	1	0	0	0	
3	3,750	0	1	1	0	0	
3	3,1250	0	1	0	1	0	
3	2,8125	5	1	0	0	1	

Here a ramp signal with an amplitude of 16V is digitized. The ADC has 4 bits and a 16V voltage range.

The red curve represents the analogue signal $x_{an}(t)$. The blue curve represents the digitized signal $x_{dig}(t)$.

We observe that the digitized signal is consistently lower in amplitude than the analogue signal. This is a **truncating ADC**.



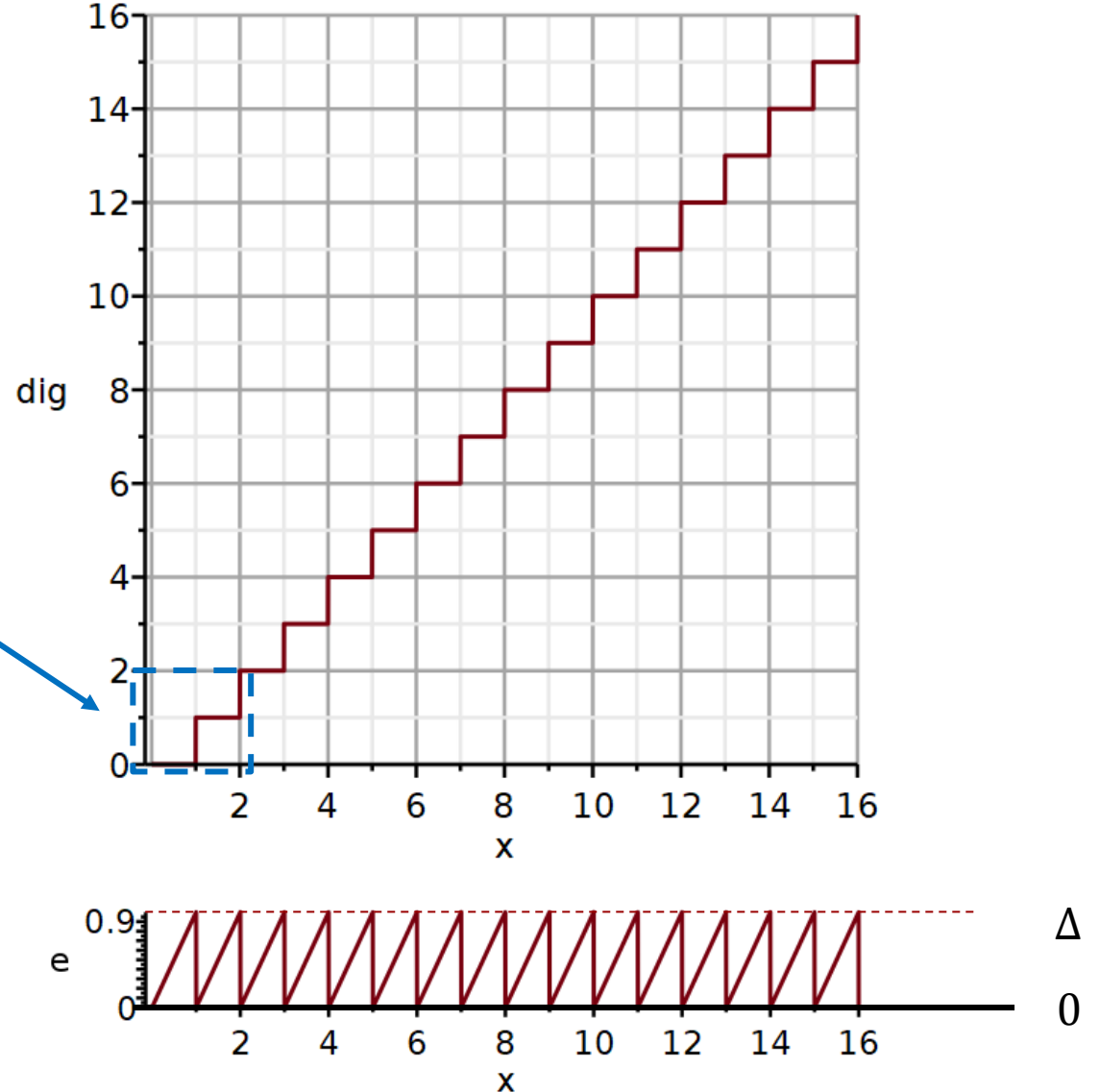
The top curve illustrates the input-output relationship of a **truncating ADC**.

When $x_{an}(t)$ climbs from 0V to 1V, the digital value $x_{dig}(t)$ remains 0V. We will define the difference as an **error signal**:

$$e(t) \stackrel{\text{def}}{=} x_{an}(t) - x_{dig}(t)$$

The lower graph illustrates the error signal $e(t)$.

The error signal has a **non-zero mean**. This error signal is referred to as quantization noise. For a truncating ADC, the **quantization noise is biased**.



The quantization error was defined as

$$e(t) \triangleq x_{an}(t) - x_{dig}(t)$$

In practice the analogue signal entering the ADC is not perfectly deterministic.

It typically consists of a desired signal plus small analogue noise:

$$x_{an}(t) = s(t) + n(t)$$

Where:

- $s(t)$ = desired signal
- $n(t)$ = analogue noise (thermal noise, amplifier noise, biological noise, interference, etc.)

The quantization error therefore becomes

$$e(t) = s(t) + n(t) - x_{dig}(t)$$

Consequence

The noise component $n(t)$ causes the signal to vary slightly relative to the quantization thresholds.

As a result, the quantization error varies from sample to sample and can be modelled as a **random variable**.

For many signals, the quantization error can be approximated as **uniformly distributed within one quantization interval**.

The probability density function then is:

$$e \sim U(0, \Delta V_{LSB})$$

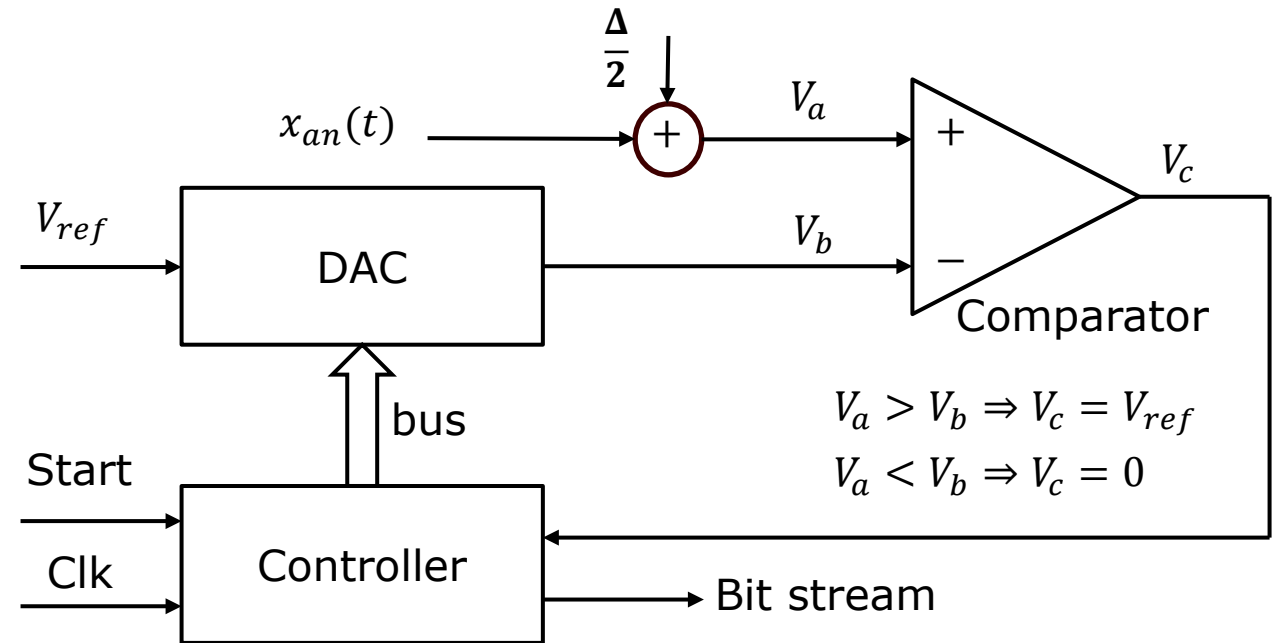
$$p(e) = \begin{cases} \frac{1}{\Delta} & 0 \leq e \leq \Delta \\ 0 & \text{otherwise} \end{cases}$$

Successive approximation, **rounding** ADC.

$\frac{\Delta}{2}$ is added to the analogue signal.

At the end of the conversion, a bit pattern is obtained corresponding to 3.125V. Hence the analogue signal of 3V is rounded up to the nearest binary value corresponding to 3.125V. The result is 0.125V higher than the analogue value.

Observation: The result can be higher than the analogue value. We will call this a **rounding ADC**.

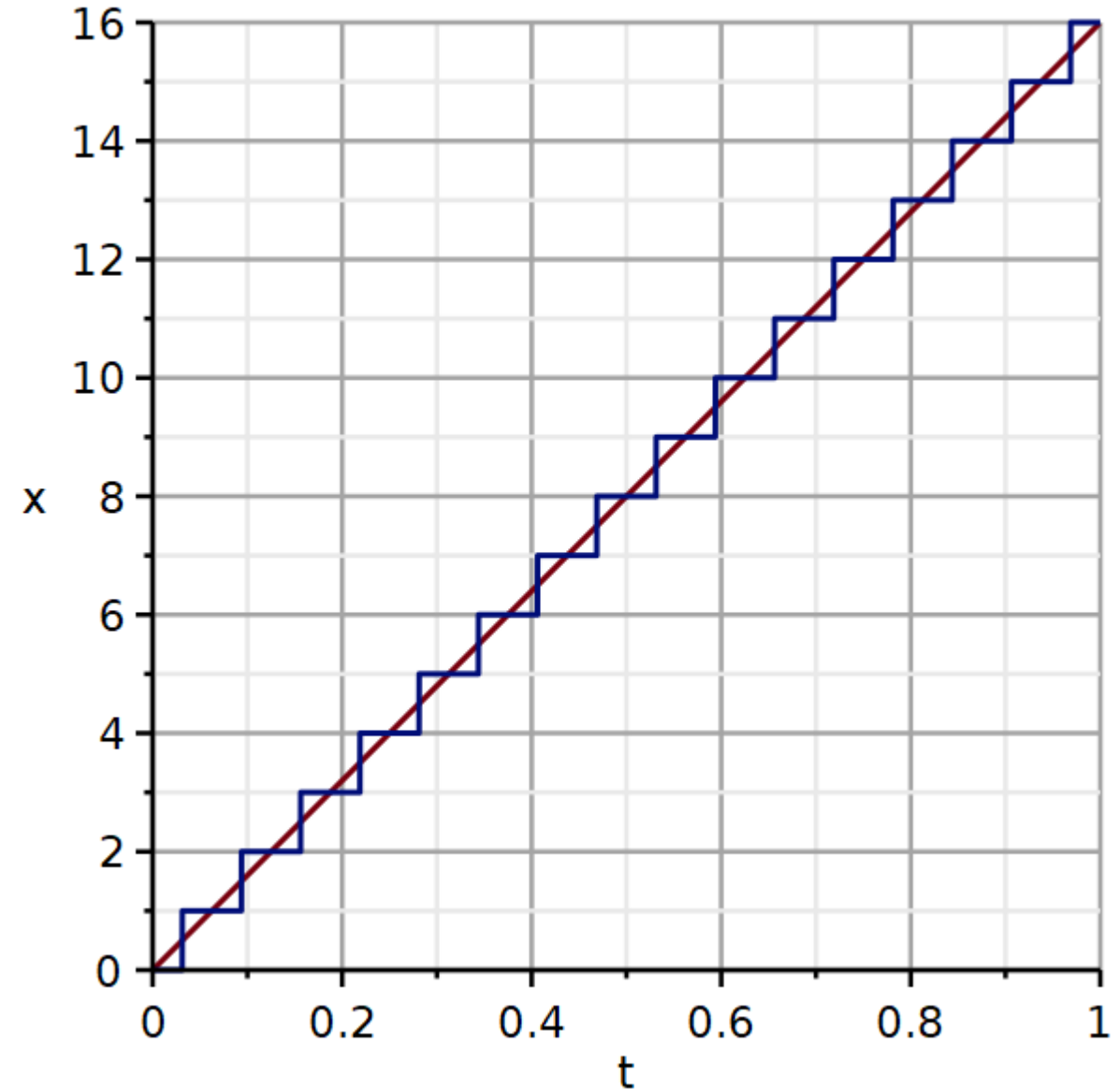


4-bit converter, $V_{ref} = 5V$

x (V)	V _a (V)	V _b (V)	V _c (V)	MSB				LSB
				3	2	1	0	0
3	3.15625	2,500	5	1	0	0	0	
3	3.15625	3,750	0	1	1	0	0	
3	3.15625	3,1250	5	1	0	1	0	
3	3.15625	3,4375	0	1	0	1	1	
3	3.15625	3,1250	5	1	0	1	0	

Here a ramp signal with an amplitude of 16V is digitized. The red curve represents the analogue signal $x_{an}(t)$. The blue curve represents the digitized signal $x_{dig}(t)$.

We observe that the digitized signal is now rounded off to nearest binary value. This is the performance seen in a **rounding ADC**.



Quantization noise of rounding ADC

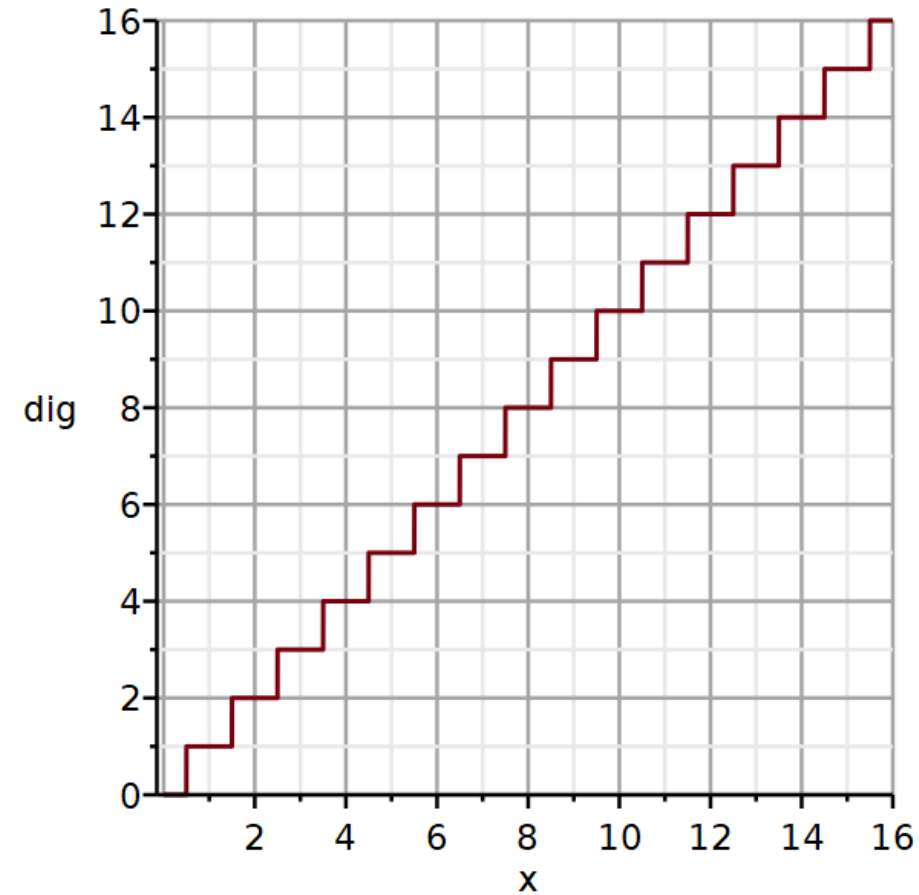
The top curve illustrates the input-output relationship of a rounding ADC.

When $x_{an}(t)$ climbs from 0V to 0.5V, the digital value $x_{dig}(t)$ remains 0V. When $x_{an}(t)$ climbs from 0.5V to 1V, the digital signal is 1V. We will define the difference as an error signal:

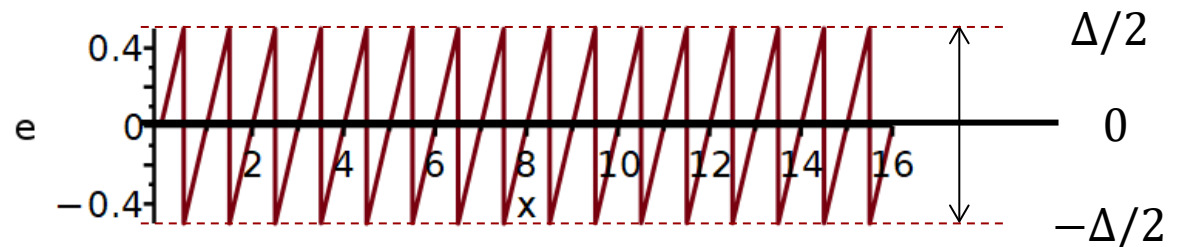
$$e(t) \stackrel{\text{def}}{=} x_{an}(t) - x_{dig}(t)$$

The lower graph illustrates the error signal $e(t)$ for a rounding ADC.

The error signal is **symmetric around zero**. For a rounding ADC the quantization noise therefore has zero mean (**unbiased**).



$$\Delta \stackrel{\text{def}}{=} V_{LSB}$$



Analogue-Digital Conversion

Quantization noise

Video

We have seen that analogue-to-digital conversion results in quantization noise. An important performance parameter for an ADC is therefore its **signal-to-noise ratio (SNR)**. SNR is usually listed in decibels. The SNR is calculated from the ratio of **rms values** of the signal and the noise.

Besides quantization noise the noise signal can also include thermal noise from resistors and 1/f noise from semiconductors.

We aim to calculate the SNR for the truncating and rounding ADC. To do this we need first to figure out how to calculate rms values of both a desired signal and the noise signal.

$$SNR = 20 \log \frac{x_{an,rms}}{n_{rms}}$$

The definition of the root mean square:

We separate the signal into a sum of its mean value and a signal with zero mean:

$$x_{an}(t) = \bar{x}_{an} + x_{an,0}(t)$$

The green integral is the mean value of a signal with zero mean.

The power of a signal is the sum of the power of the mean and the power of the zero-mean component. The latter is the variance of the signal.

$$x_{an,rms} \stackrel{\text{def}}{=} \sqrt{\frac{1}{T} \int_{-T/2}^{T/2} x_{an}^2(t) dt}$$

$$x_{an,rms}^2 = \frac{1}{T} \int_{-T/2}^{T/2} (\bar{x}_{an} + x_{an,0}(t))^2 dt$$

$$x_{an,rms}^2 = \frac{1}{T} \int_{-T/2}^{T/2} (\bar{x}_{an}^2 + 2\bar{x}_{an}x_{an,0}(t) + x_{an,0}^2(t)) dt$$

$$= \bar{x}_{an}^2 + 2\bar{x}_{an} \underbrace{\frac{1}{T} \int_{-T/2}^{T/2} x_{an,0}(t) dt}_{=0} + \frac{1}{T} \int_{-T/2}^{T/2} x_{an,0}^2(t) dt$$

$$= \bar{x}_{an}^2 + \sigma_{x_{an}}^2$$

$$x_{an,rms} = \sqrt{\bar{x}_{an}^2 + \sigma_{x_{an}}^2}$$

For the sake of illustration, we will assume that the analogue signal (desired signal) is a sinusoidal signal with a constant offset:

$$x_{an}(t) = \frac{A}{2} + \frac{A}{2} \sin \omega t$$

The full voltage range of the ADC is:

For a sinusoidal signal with a peak-to-peak amplitude A which is only a fraction α of the ADC voltage range, we have:

Its rms value is:

$$x_{an,rms} = \sqrt{\frac{1}{T} \int_{-T/2}^{T/2} x_{an}^2(t) dt}$$

$$x_{an,rms} = \sqrt{\frac{A^2}{4} + \frac{A^2}{8}} = \sqrt{\frac{3A^2}{8}} = A \sqrt{\frac{3}{8}} \approx 0.612A$$

$$\text{Full scale} = 2^N \times \Delta V_{LSB}$$

$$A = \alpha 2^N \times \Delta V_{LSB}$$

$$x_{an,rms} = \sqrt{\frac{3}{8}} \alpha 2^N \times \Delta V_{LSB}$$

To derive the rms value of the quantization noise we need to calculate the **mean and variance of a random variable X** .

For a random variable with uniform probability distribution from a to b .

$$n_{rms} = \sqrt{\bar{n}^2 + \sigma_n^2}$$

$$\mu = \frac{a+b}{2} \quad \sigma^2 = \frac{(b-a)^2}{12}$$

For biased quantization noise in the interval $[0, \Delta]$:

$$\bar{n} = \frac{\Delta}{2} \quad \sigma_n^2 = \frac{\Delta^2}{12} \quad n_{rms,b} = \sqrt{\left(\frac{\Delta}{2}\right)^2 + \frac{\Delta^2}{12}} = \frac{\Delta}{\sqrt{3}}$$

For unbiased quantization noise in the interval $[-\Delta/2, \Delta/2]$:

$$\bar{n} = 0 \quad \sigma_n^2 = \frac{\Delta^2}{12} \quad n_{rms,ub} = \sqrt{\frac{\Delta^2}{12}} = \frac{\Delta}{\sqrt{12}}$$

The rms value of the biased noise is 2 times that of the unbiased noise.

$$\frac{n_{rms,b}}{n_{rms,ub}} = 2$$

If several ADC conversions are made in rapid succession, and the average value is calculated from the set of samples, then the analogue signal ($s(t)$) may not have changed significantly in the short time duration, but the quantization noise $n(t)$ may be reduced due to cancelation of noise samples with opposite signs.

Hence there is a possibility of suppressing noise by using the average of a set of samples obtained at a higher sampling frequency. This technique is called **oversampling**.

From statistics we use the results of averaging a random variable X with mean μ and variance σ^2 :

$$Y = \frac{1}{M} \sum_{i=1}^M X_i$$

We observe that averaging does not change the mean value:

$$E[Y] = \mu$$

However, the variance is reduced in proportion to the number of averages:

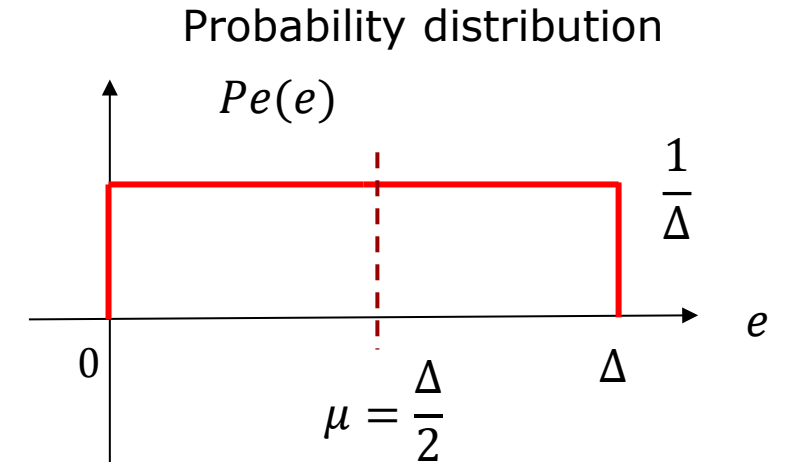
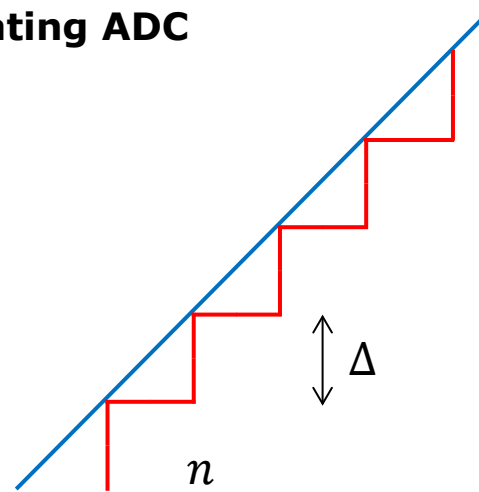
$$\text{Var}(Y) = \frac{\sigma^2}{M}$$

If the signal has a non-zero mean, the benefit of averaging will be limited.

$$Y_{rms} = \sqrt{\mu^2 + \frac{\sigma^2}{M}}$$

RMS value of quantization noise for truncating ADC

Using these results for the **truncating ADC**, we see that averaging has limited effect on reducing the noise due to the bias (non-zero mean).



$$X_{rms,M} = \sqrt{\mu^2 + \frac{\sigma^2}{M}}$$

$$\mu^2 = \left(\frac{\Delta}{2}\right)^2$$

$$\sigma^2 = \frac{(b-a)^2}{12} = \frac{(\Delta V_{LSB} - 0)^2}{12} = \frac{\Delta^2}{12}$$

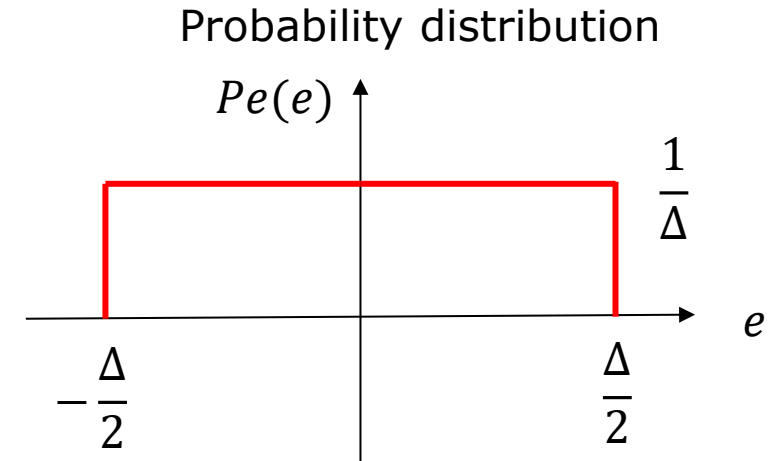
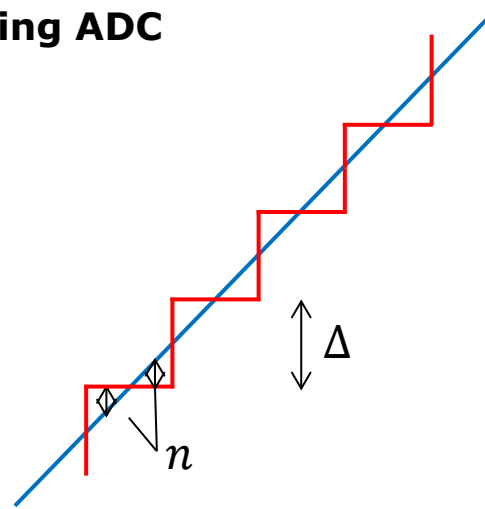
Observation:

Averaging samples from a truncating ADC reduces the variance, but not the mean.

$$n_{rms,M} = \sqrt{\mu^2 + \frac{\sigma^2}{M}} = \sqrt{\left(\frac{\Delta}{2}\right)^2 + \left(\frac{\Delta}{12M}\right)^2}$$

RMS value of quantization noise for rounding ADC

The quantization noise from a **rounding ADC** has zero mean value, hence oversampling is more effective in reducing quantization noise.



$$X_{rms,M} = \sqrt{\mu^2 + \frac{\sigma^2}{M}}$$

$$\mu^2 = 0$$

$$\sigma^2 = \frac{(b-a)^2}{12} = \frac{\left(\frac{\Delta}{2} - \frac{-\Delta}{2}\right)^2}{12} = \frac{\Delta^2}{12}$$

$$n_{rms,M} = \sqrt{\frac{\Delta^2}{12M}}$$

Observation:

Averaging samples from a rounding ADC reduces the variance and thus the quantization noise.

SNR of a truncating ADC:

$$SNR = 20 \log \frac{x_{an,rms}}{n_{rms,M}} = 20 \log \frac{\sqrt{\frac{3}{8}} \alpha 2^N \times \Delta}{\sqrt{\left(\frac{\Delta}{2}\right)^2 + \left(\frac{\Delta}{\sqrt{12M}}\right)^2}} = 20 \log \frac{\sqrt{\frac{3}{8}} \alpha 2^N}{\sqrt{\frac{1}{4} + \frac{1}{12M}}}$$

Separating out terms, we can better judge the significance of each factor.

$$SNR = 20 \log(2^N) + 20 \log(\alpha) + 20 \log\left(\sqrt{\frac{3}{8}}\right) - 20 \log \sqrt{\frac{1}{4} + \frac{1}{12M}}$$

$$= N20 \log(2) + 20 \log(\alpha) - 4.26\text{dB} - 20 \log \sqrt{\frac{1}{4} + \frac{1}{12M}}$$

$$\approx 6.02N + 20 \log(\alpha) - 4.26\text{dB} - 20 \log \sqrt{\frac{1}{4} + \frac{1}{12M}}$$

SNR of a rounding ADC:

$$SNR = 20 \log \frac{x_{an,rms}}{n_{rms,M}} = 20 \log \frac{\sqrt{\frac{3}{8}} \alpha 2^N \times \Delta}{\sqrt{\left(\frac{\Delta}{\sqrt{12M}}\right)^2}} = 20 \log \frac{\sqrt{\frac{3}{8}} \alpha 2^N}{\sqrt{\frac{1}{12M}}}$$

$$SNR = 20 \log(2^N) + 20 \log(\alpha) + 20 \log\left(\sqrt{\frac{3}{8}}\right) - 20 \log \sqrt{\frac{1}{12M}}$$

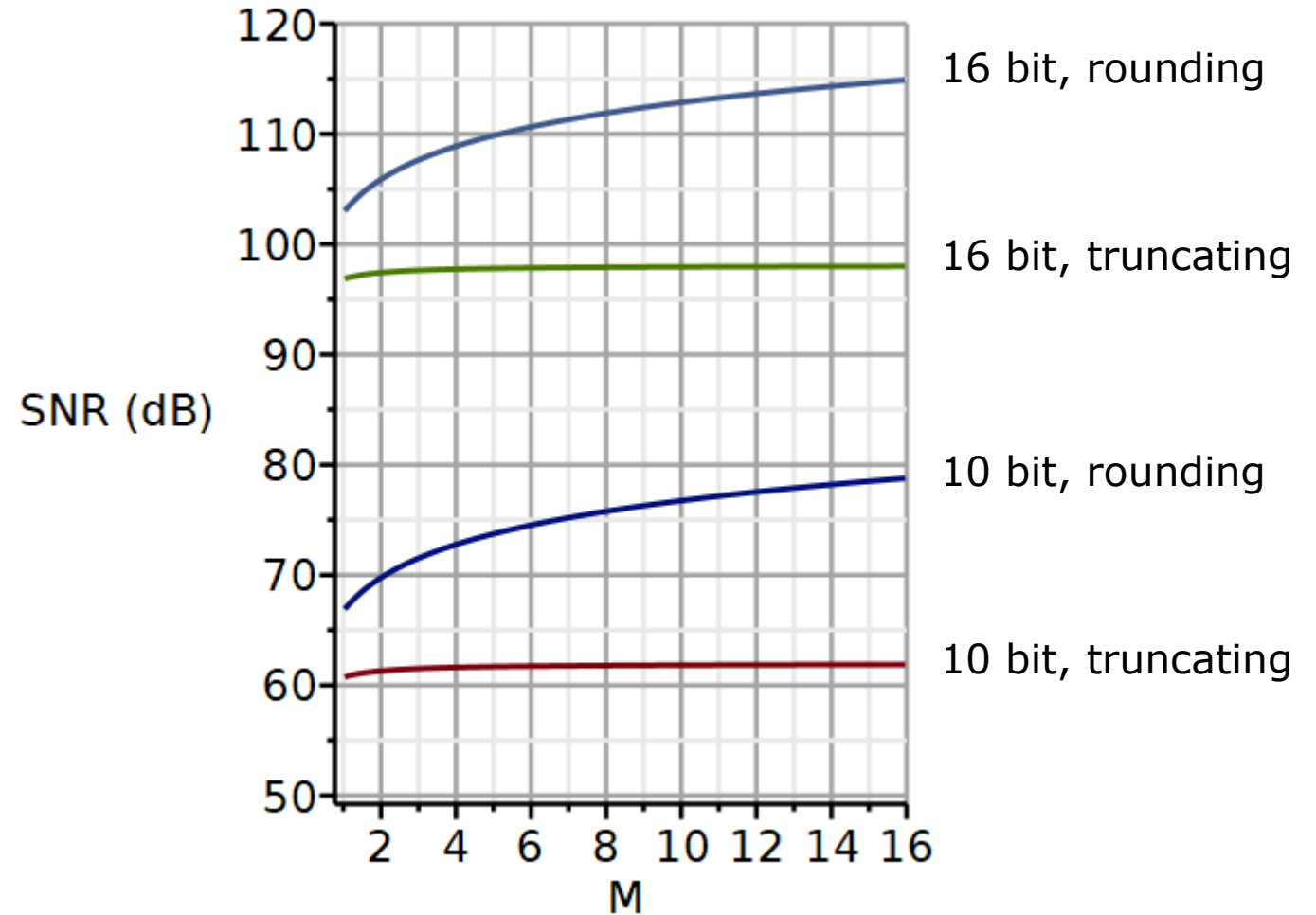
$$= N20 \log(2) + 20 \log(\alpha) - 4.26\text{dB} + 20 \log \sqrt{12} + 20 \log \sqrt{M}$$

$$\approx 6.02N + 20 \log(\alpha) + 20 \log \sqrt{M} - 4.26\text{dB} + 10.79\text{dB}$$

Comparing SNR for truncating and rounding ADCs

Observation:

There is essentially no improvement in SNR due to quantization noise by oversampling, if the ADC is of the truncating type.



We see that the SNR increases 6dB per bit.

$$SNR \approx 6.02N + 20 \log(\alpha) + 20 \log \sqrt{M} - 4.26\text{dB} + 10.79\text{dB}$$

We can then figure out how many averages we need to improve the SNR equivalent to one bit:

$$20 \log \sqrt{M} = 6.02 \Rightarrow M = 4$$

The number of averages performed can be expressed as:

$$M = 4^x \quad x: \text{The number of bits gained by averaging.}$$

$$SNR \approx 6.02N + 20 \log(\alpha) + 20 \log \sqrt{4^x} - 4.26\text{dB} + 10.79\text{dB}$$

$$SNR \approx 6.02N + 20 \log(\alpha) + 20 \log 4^{x/2} - 4.26\text{dB} + 10.79\text{dB}$$

$$SNR \approx 6.02N + 20 \log(\alpha) + x \times 20 \log 2 - 4.26\text{dB} + 10.79\text{dB}$$

$$SNR \approx 6.02N + 20 \log(\alpha) + x \times 6.02\text{dB} - 4.26\text{dB} + 10.79\text{dB}$$

Example:

we have a 10-bit rounding ADC, but need a SNR equivalent of a 12-bit ADC, we must obtain a 2-bit improvement in SNR. Hence the number of averages required is:

$$SNR \approx 6.02N + 20 \log(\alpha) + x \times 6.02\text{dB} - 4.26\text{dB} + 10.79\text{dB}$$

$$M = 4^x = 4^2 = 16$$

Checking with the graphs on the previous slide, we confirm a 12 dB (2-bit improvement) by oversampling by 16.

If the full voltage range of the ADC is not utilized, not all bits are put to use, and the SNR will be less than optimal.

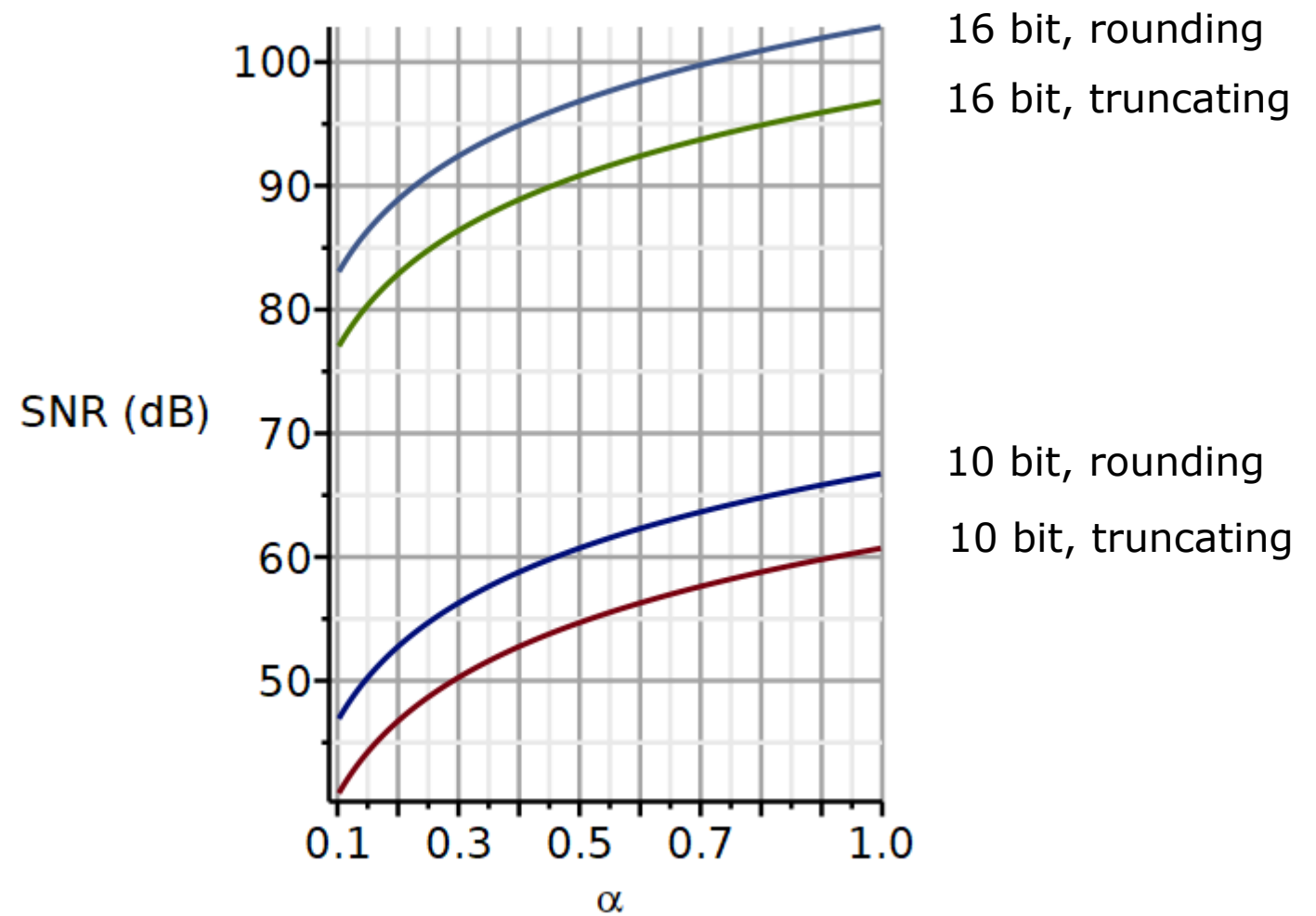
Here are shown the dependence of SNR on the parameter α which represents the fractional utilization of the full voltage range.

No oversampling is used here.

The SNR improves by $20 \log(\alpha)$.

Hence a doubling in amplification will improve the SNR by 6dB $\sim$ 1bit.

A factor 10 increase in amplification will improve the SNR by 20dB $\sim$ 3.3bits.



The fewer bits you have in your ADC, the more important it is to provide adequate amplification. On the other hand, too high amplification will cause the amplified signal to exceed the voltage range of the ADC. This will cause **clipping**, as the digital level cannot exceed the full range.

The effects of clipping are serious and renders the digitized data useless. If the maximum amplitude of a signal is hard to predict, it is necessary to allow some **head room** to avoid clipping.

If we replace a 10-bit ADC with a 16-bit ADC, we need to decide if we want to use all the 6 extra bits to reduce code width, or if some of the extra bits should be used to increase head room.

Matching antialiasing filter to ADC

Video

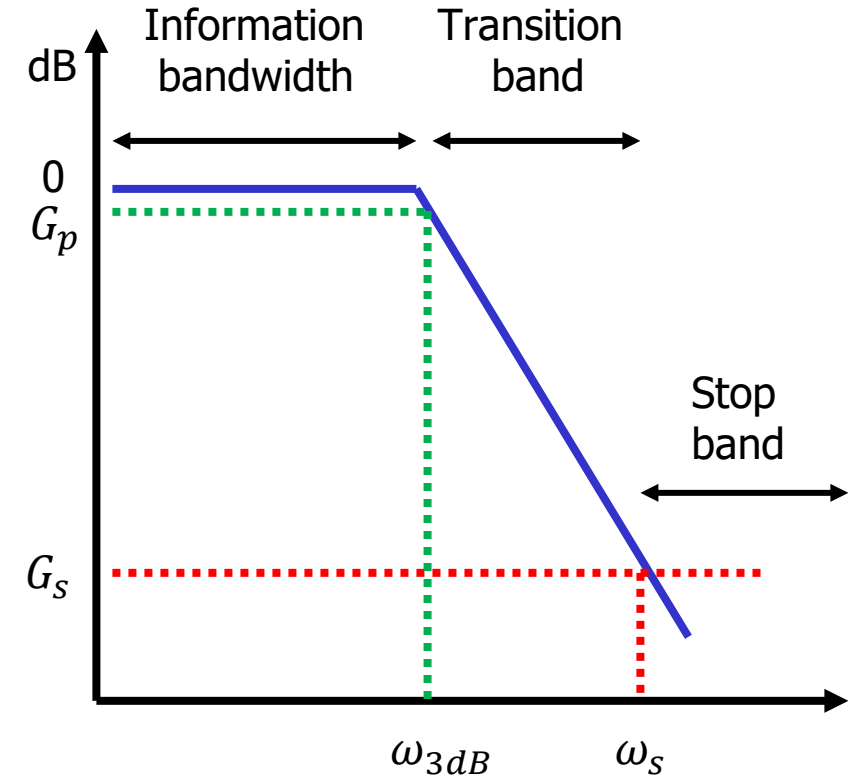
From our analysis of the ADC performance, we have learned that

1. Inadequate sampling frequency will increase the negative consequences of aliasing.
2. Inadequate bits will increase quantization noise.

Because all practical signals have **finite duration**, their frequency spectrum has **infinite bandwidth**. No matter how high the sampling frequency is, some level of aliasing will occur.

An antialiasing lowpass filter can attenuate the energy in frequency ranges outside the information bandwidth. Adequate filter order will therefore attenuate the aliased frequency components to a level that can be considered insignificant.

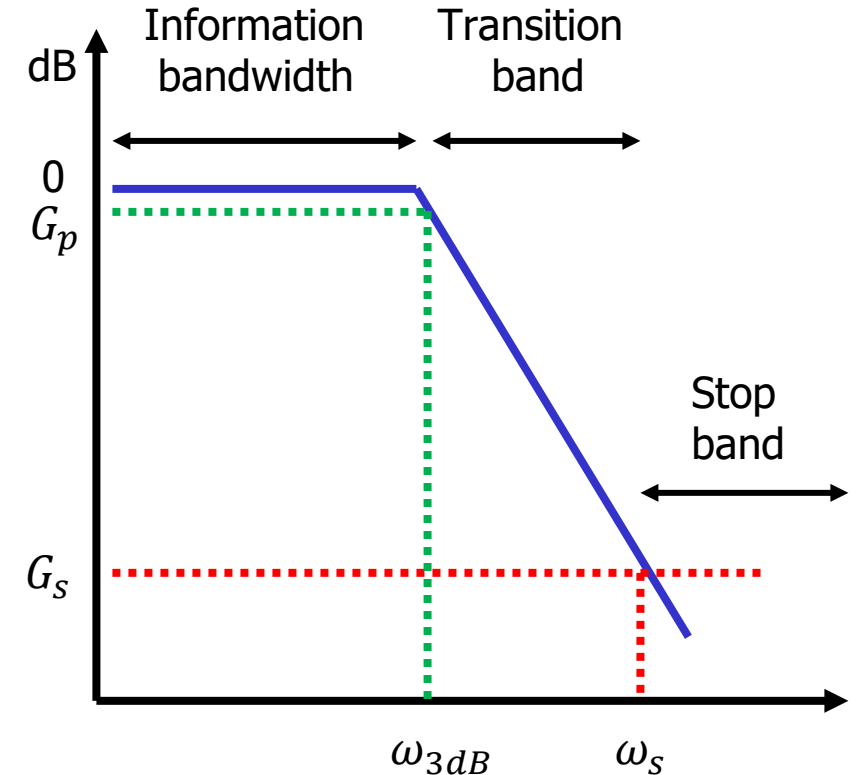
In the figure shown here, the desired level of attenuation is denoted G_s , referring to the gain at the start of the stop band.



Nyquist sampling theorem states that the sampling frequency should be at least two times the highest frequency of significant magnitude.

We can then make the choice that the highest frequency of significant amplitude should define the start of the stop band of the antialiasing filter. In other words, we set:

$$\omega_s = \frac{\omega_{Fs}}{2}$$



Another important decision is the level of attenuation G_s at $\omega = \omega_s$. This is the amplitude of the frequencies folded about $\omega_{FS}/2$.

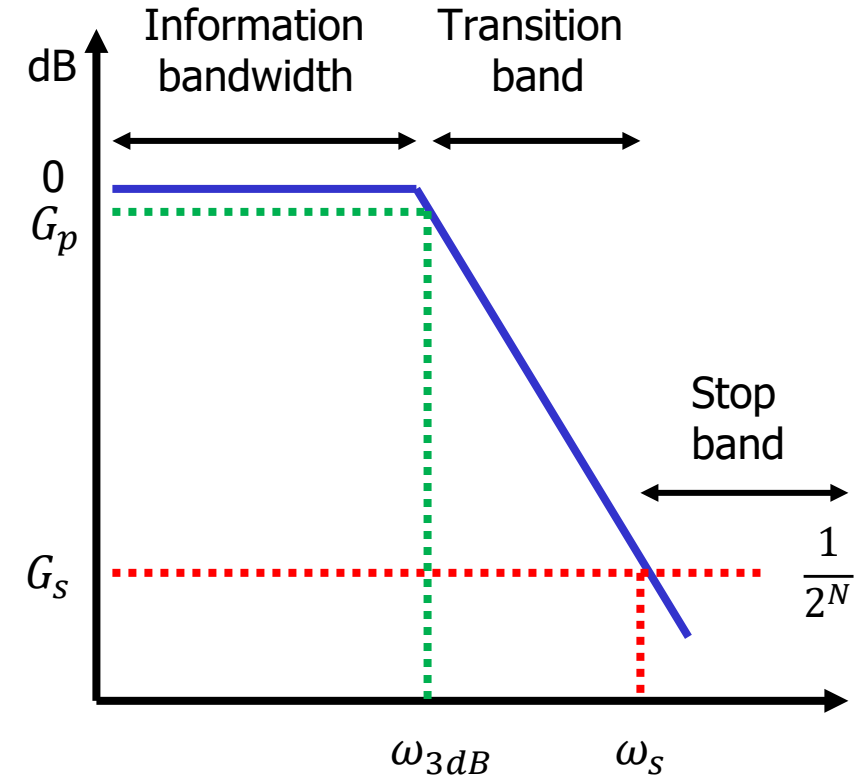
We could fix state that the aliased frequencies should be attenuated by at least 1000 (60 dB).

For a given level of attenuation, we can ask how many bits can be toggled by the aliased frequencies.

A 12-bit ADC will toggle the LSB if the signal exceeds -72dB.

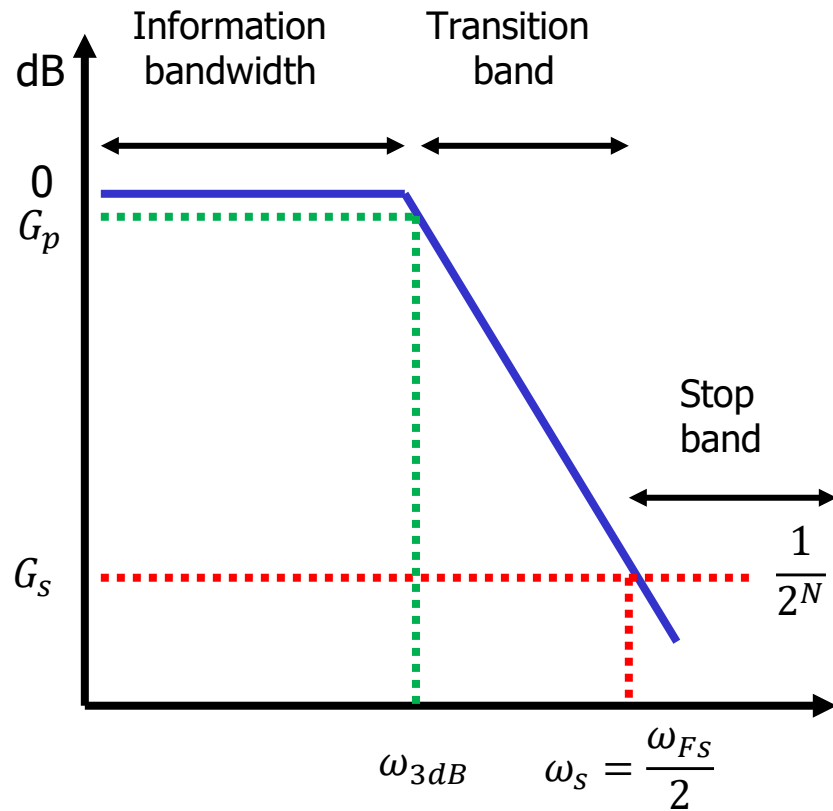
So, if the level of attenuation is 60dB at $\omega_{FS}/2$, we should expect that aliased frequency components will have enough amplitude to toggle the 2 LSB of the ADC.

Let us try and be extreme and insist that the filter attenuation equals $\Delta = V_{LSB}$:



$$G_s = 20 \log \frac{1}{2^N}$$

Ratio between 3dB cut-off and sampling frequency



Setting the stop band frequency equal to half the sampling frequency:

$$|H(j\omega_s)| \stackrel{\text{def}}{=} \frac{1}{\sqrt{1 + \left(\frac{\omega_s}{\omega_{3dB}}\right)^{2n}}} = \frac{1}{2^N} \quad \frac{1}{1 + \left(\frac{\omega_s}{\omega_{3dB}}\right)^{2n}} = \frac{1}{2^{2N}}$$

$$1 + \left(\frac{\omega_s}{\omega_{3dB}}\right)^{2n} = 2^{2N} \quad \left(\frac{\omega_s}{\omega_{3dB}}\right)^{2n} = 2^{2N} - 1$$

$$N \geq 8 \Rightarrow \text{error} \leq \frac{|(2^{16} - 1) - 2^{16}|}{(2^{16} - 1)} \approx 1.53 \times 10^{-5}$$

$$\left(\frac{\omega_s}{\omega_{3dB}}\right)^{2n} \approx 2^{2N} \Rightarrow \frac{\omega_s}{\omega_{3dB}} \approx 2^{\frac{N}{n}}$$

$$\omega_s = \frac{\omega_{FS}}{2} \Rightarrow$$

$$\boxed{\frac{\omega_{FS}}{\omega_{3dB}} = 2^{\frac{N}{n}+1}}$$

Ratios between 3dB cut-off and sampling frequency

$$\frac{\omega_{Fs}}{\omega_{3dB}} = 2^{\frac{N}{n}+1}$$

n	N	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144 (2 ^N bits)
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18 bits
1		4.00	8.00	16.00	32.00	64.00	128.00	256.00	512.00	1024.00	2048.00	4096.00	8192.00	16384.00	32768.00	65536.00	131072.00	262144.00	524288.00
2		2.83	4.00	5.66	8.00	11.31	16.00	22.63	32.00	45.25	64.00	90.51	128.00	181.02	256.00	362.04	512.00	724.08	1024.00
3		2.52	3.17	4.00	5.04	6.35	8.00	10.08	12.70	16.00	20.16	25.40	32.00	40.32	50.80	64.00	80.63	101.59	128.00
4		2.38	2.83	3.36	4.00	4.76	5.66	6.73	8.00	9.51	11.31	13.45	16.00	19.03	22.63	26.91	32.00	38.05	45.25
5		2.30	2.64	3.03	3.48	4.00	4.59	5.28	6.06	6.96	8.00	9.19	10.56	12.13	13.93	16.00	18.38	21.11	24.25
6		2.24	2.52	2.83	3.17	3.56	4.00	4.49	5.04	5.66	6.35	7.13	8.00	8.98	10.08	11.31	12.70	14.25	16.00
7		2.21	2.44	2.69	2.97	3.28	3.62	4.00	4.42	4.88	5.38	5.94	6.56	7.25	8.00	8.83	9.75	10.77	11.89
8		2.18	2.38	2.59	2.83	3.08	3.36	3.67	4.00	4.36	4.76	5.19	5.66	6.17	6.73	7.34	8.00	8.72	9.51
9		2.16	2.33	2.52	2.72	2.94	3.17	3.43	3.70	4.00	4.32	4.67	5.04	5.44	5.88	6.35	6.86	7.41	8.00
10		2.14	2.30	2.46	2.64	2.83	3.03	3.25	3.48	3.73	4.00	4.29	4.59	4.92	5.28	5.66	6.06	6.50	6.96

In this table, the rows 1 – 10 is the filter order n . The columns represent the number of bits N in the ADC.

Example: Fourth order Butterworth lowpass filter, 10-bit ADC.

$\frac{\omega_{Fs}}{\omega_{3dB}} = 11.31$. Hence if the 3dB cut-off frequency is 100Hz, the sampling frequency needs to be 1131Hz.

At this sampling frequency, the aliased signal frequencies have been attenuated by a factor 1024 or more. This is just below the LSB amplitude level and will be non-detectable for a 10-bit ADC.

Ratios between 3dB cut-off and sampling frequency

$$\frac{\omega_{Fs}}{\omega_{3dB}} = 2^{\frac{N}{n}+1}$$

n	N	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144 (2^bits)
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18 bits
1		4.00	8.00	16.00	32.00	64.00	128.00	256.00	512.00	1024.00	2048.00	4096.00	8192.00	16384.00	32768.00	65536.00	131072.00	262144.00	524288.00
2		2.83	4.00	5.66	8.00	11.31	16.00	22.63	32.00	45.25	64.00	90.51	128.00	181.02	256.00	362.04	512.00	724.08	1024.00
3		2.52	3.17	4.00	5.04	6.35	8.00	10.08	12.70	16.00	20.16	25.40	32.00	40.32	50.80	64.00	80.63	101.59	128.00
4		2.38	2.83	3.36	4.00	4.76	5.66	6.73	8.00	9.51	11.31	13.45	16.00	19.03	22.63	26.91	32.00	38.05	45.25
5		2.30	2.64	3.03	3.48	4.00	4.59	5.28	6.06	6.96	8.00	9.19	10.56	12.13	13.93	16.00	18.38	21.11	24.25
6		2.24	2.52	2.83	3.17	3.56	4.00	4.49	5.04	5.66	6.35	7.13	8.00	8.98	10.08	11.31	12.70	14.25	16.00
7		2.21	2.44	2.69	2.97	3.28	3.62	4.00	4.42	4.88	5.38	5.94	6.56	7.25	8.00	8.83	9.75	10.77	11.89
8		2.18	2.38	2.59	2.83	3.08	3.36	3.67	4.00	4.36	4.76	5.19	5.66	6.17	6.73	7.34	8.00	8.72	9.51
9		2.16	2.33	2.52	2.72	2.94	3.17	3.43	3.70	4.00	4.32	4.67	5.04	5.44	5.88	6.35	6.86	7.41	8.00
10		2.14	2.30	2.46	2.64	2.83	3.03	3.25	3.48	3.73	4.00	4.29	4.59	4.92	5.28	5.66	6.06	6.50	6.96

Example: Fourth order Butterworth lowpass filter, 10-bit ADC.

If we are willing to accept the signal level only to be reduced by a factor 256 (~8 bits), then we can sample with 800 Hz and the signal level will toggle the two LSB of a 10-bit ADC.

Ratio between 3dB cut-off and sampling frequency

$$\frac{\omega_{Fs}}{\omega_{3dB}} = 2^{\frac{N}{n}+1}$$

		2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144 (2^bits)	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	bits
1		4.00	8.00	16.00	32.00	64.00	128.00	256.00	512.00	1024.00	2048.00	4096.00	8192.00	16384.00	32768.00	65536.00	131072.00	262144.00	524288.00	
2		2.83	4.00	5.66	8.00	11.31	16.00	22.63	32.00	45.25	64.00	90.51	128.00	181.02	256.00	362.04	512.00	724.08	1024.00	
3		2.52	3.17	4.00	5.04	6.35	8.00	10.08	12.70	16.00	20.16	25.40	32.00	40.32	50.80	64.00	80.63	101.59	128.00	
4		2.38	2.83	3.36	4.00	4.76	5.66	6.73	8.00	9.51	11.31	13.45	16.00	19.03	22.63	26.91	32.00	38.05	45.25	
5		2.30	2.64	3.03	3.48	4.00	4.59	5.28	6.06	6.96	8.00	9.19	10.56	12.13	13.93	16.00	18.38	21.11	24.25	
6		2.24	2.52	2.83	3.17	3.56	4.00	4.49	5.04	5.66	6.35	7.13	8.00	8.98	10.08	11.31	12.70	14.25	16.00	
7		2.21	2.44	2.69	2.97	3.28	3.62	4.00	4.42	4.88	5.38	5.94	6.56	7.25	8.00	8.83	9.75	10.77	11.89	
8		2.18	2.38	2.59	2.83	3.08	3.36	3.67	4.00	4.36	4.76	5.19	5.66	6.17	6.73	7.34	8.00	8.72	9.51	
9		2.16	2.33	2.52	2.72	2.94	3.17	3.43	3.70	4.00	4.32	4.67	5.04	5.44	5.88	6.35	6.86	7.41	8.00	
10		2.14	2.30	2.46	2.64	2.83	3.03	3.25	3.48	3.73	4.00	4.29	4.59	4.92	5.28	5.66	6.06	6.50	6.96	
order																				

Assume a 12-bit ADC and a 4th order lowpass filter.

We insist that: $\frac{\omega_{Fs}}{\omega_{3dB}} \leq 4$

Now the filter will only attenuate the aliased frequency to the LSB of a 4-bit ADC, i.e., by a factor of 16.

Ratio between 3dB cut-off and sampling frequency

$$\frac{\omega_{Fs}}{\omega_{3dB}} = 2^{\frac{N}{n}+1}$$

	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144 (2^bits)
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18 bits
1	4.00	8.00	16.00	32.00	64.00	128.00	256.00	512.00	1024.00	2048.00	4096.00	8192.00	16384.00	32768.00	65536.00	131072.00	262144.00	524288.00
2	2.83	4.00	5.66	8.00	11.31	16.00	22.63	32.00	45.25	64.00	90.51	128.00	181.02	256.00	362.04	512.00	724.08	1024.00
3	2.52	3.17	4.00	5.04	6.35	8.00	10.08	12.70	16.00	20.16	25.40	32.00	40.32	50.80	64.00	80.63	101.59	128.00
4	2.38	2.83	3.36	4.00	4.76	5.66	6.73	8.00	9.51	11.31	13.45	16.00	19.03	22.63	26.91	32.00	38.05	45.25
5	2.30	2.64	3.03	3.48	4.00	4.59	5.28	6.06	6.96	8.00	9.19	10.56	12.13	13.93	16.00	18.38	21.11	24.25
6	2.24	2.52	2.83	3.17	3.56	4.00	4.49	5.04	5.66	6.35	7.13	8.00	8.98	10.08	11.31	12.70	14.25	16.00
7	2.21	2.44	2.69	2.97	3.28	3.62	4.00	4.42	4.88	5.38	5.94	6.56	7.25	8.00	8.83	9.75	10.77	11.89
8	2.18	2.38	2.59	2.83	3.08	3.36	3.67	4.00	4.36	4.76	5.19	5.66	6.17	6.73	7.34	8.00	8.72	9.51
9	2.16	2.33	2.52	2.72	2.94	3.17	3.43	3.70	4.00	4.32	4.67	5.04	5.44	5.88	6.35	6.86	7.41	8.00
10	2.14	2.30	2.46	2.64	2.83	3.03	3.25	3.48	3.73	4.00	4.29	4.59	4.92	5.28	5.66	6.06	6.50	6.96

We insist that: $\frac{\omega_{Fs}}{\omega_{3dB}} \leq 4$

If we want the attenuation factor to be 100 (40dB ~ 7 bits), then we may decide to use a 7th order filter.

We must conclude that the constraint of sampling at 4 times the 3dB cut-off frequency is quite severe. It either demands a high-order filter or forces us to accept a poor attenuation of aliased frequencies.

Determining the filter order, Lathi 7.5

Regardless of the criteria used to define the passband gain G_p and the stopband gain G_s , a suitable filter order n must be determined, to obtain the desired passband and stopband attenuations.

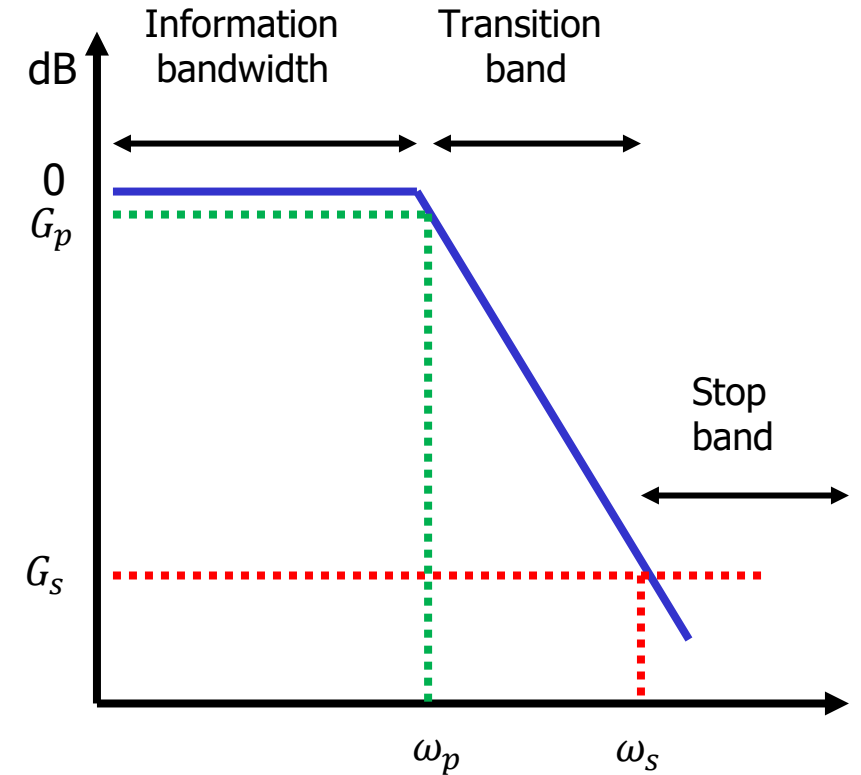
Inserting the passband and stopband frequencies into the amplitude spectrum of the Butterworth filter we get:

$$G_p = 20 \log_{10} \frac{1}{\sqrt{1 + (\omega_p/\omega_c)^{2n}}}$$

$$= -10 \log \left[1 + (\omega_p/\omega_c)^{2n} \right] : \text{Passband gain}$$

$$G_s = 20 \log_{10} \frac{1}{\sqrt{1 + (\omega_s/\omega_c)^{2n}}}$$

$$= -10 \log_{10} [1 + (\omega_s/\omega_c)^{2n}] : \text{Stopband gain}$$



Determining the filter order, Lathi 7.5

Solving for the filter order:

$$n = \frac{\log_{10} \left(\frac{10^{-\frac{G_s}{10}} - 1}{10^{-\frac{G_p}{10}} - 1} \right)}{2 \log_{10} \left(\frac{\omega_s}{\omega_p} \right)}$$

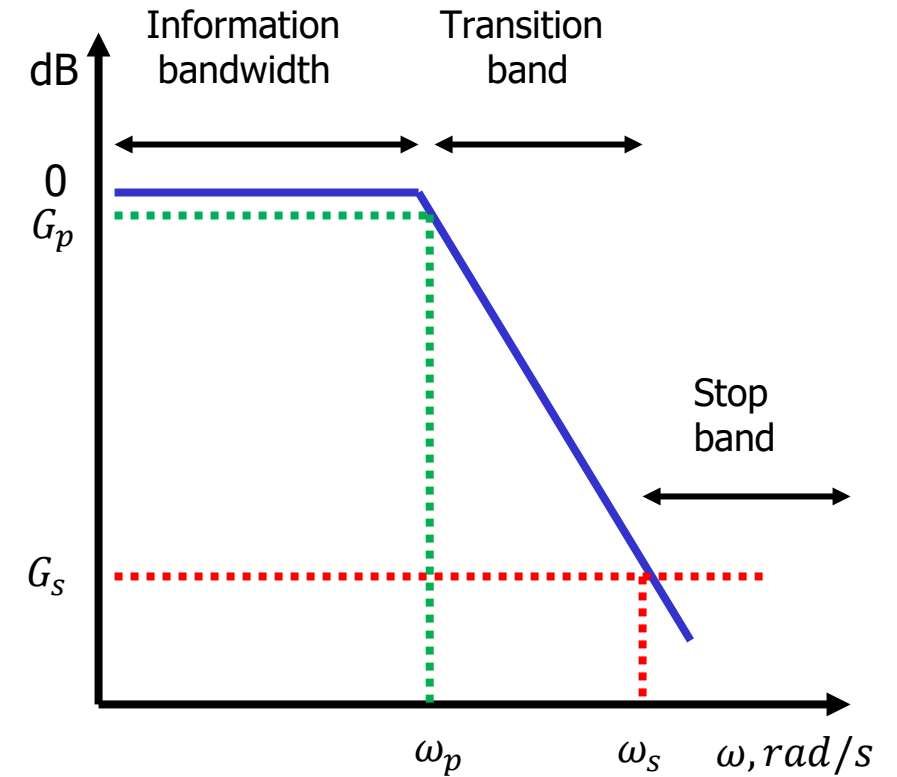
We also need to define ω_c when n is rounded off to an integer:

$$\omega_c = \frac{\omega_p}{[10^{-G_p/10} - 1]^{1/2n}}$$

$$G(s) = \frac{1}{1 + \left(\frac{s}{j\omega_c} \right)^{2n}}$$

After designing the frequency normalized filter of order n , a frequency scaling must be done:

$$K_F = \frac{\omega_c}{1}$$



Example:

A signal has information bandwidth from 0 – 150Hz. It will be sampled with a 16-bit ADC with a sampling frequency of 3kHz.

The anti-aliasing filter is a Butterworth lowpass filter.

At 150Hz the signal must not be attenuated more than 3dB.

The anti-aliasing filter should attenuate the signal to a level equal to or below LSB at half the sampling frequency.

Calculate the minimum filter order and the resulting cut-off frequency in radians.

First step is to define passband and stopband:

$$\text{Pass band: } G_p = -3\text{dB}$$

$$\omega_p = 942.5\text{rad/s}$$

$$\text{Stop band: } \omega_s = \frac{\omega_{Fs}}{2} = 9425\text{rad/s}$$

$$G_s = -96\text{dB} \approx 16 \text{ bit} \times (-6\text{dB})$$

Example continued

Calculate the minimum filter order and the resulting cut-off frequency in radians:

Pass band: $G_p = -3\text{dB}$

$$\omega_p = 942.5\text{rad/s}$$

Stop band: $G_s = -96\text{dB} \approx 16 \text{ bit} \times (-6\text{dB})$

$$\omega_s = 9425\text{rad/s}$$

We would need a 5th order filter if we would like to attenuate by 96 dB within one decade.

The frequency normalized filter must be frequency scaled by a factor:

$$n = \frac{\log\left(\frac{10^{-\frac{G_s}{10}} - 1}{10^{-\frac{G_p}{10}} - 1}\right)}{2 \log\left(\frac{\omega_s}{\omega_p}\right)}$$

$$n = \frac{\log\left(\frac{10^{-\frac{-96}{10}} - 1}{10^{-\frac{-3}{10}} - 1}\right)}{2 \log\left(\frac{9425}{942.5}\right)} = \frac{9.602}{2} = 4.8 \approx 5$$

$$\omega_c = \frac{\omega_p}{[10^{-G_p/10} - 1]^{1/2n}}$$

$$\omega_c = \frac{942.5}{[10^{0.3} - 1]^{1/10}} = 942.9 \text{ rad/s}$$

$$K_F = \frac{\omega_c}{1 \text{ rad/s}} = 942.9$$

Problems

Problem 1 Matching filter to ADC**Part 1:**

A signal has information bandwidth from 0 – 500Hz. It will be sampled with a 12-bit ADC.

The anti-aliasing filter is a Butterworth lowpass filter.

At 500Hz the signal must not be attenuated more than 3dB.

The anti-aliasing filter should attenuate the signal to a level equal to -72 dB at half the sampling frequency.

Part 2:

Calculate the minimum filter order required if the sampling frequency is 6kHz.

We only have a 4th order Butterworth lowpass filter at our disposal.

Determine what sampling frequency F_s is required to obtain -72dB filter gain at $F_s/2$.

Part 3:

Conflicting design specifications defines a max sampling frequency of 4kHz.

What is the filter attenuation at $F_s/2$?

How many bits are toggled at this frequency?

Solutions

Problem 1 Determining the filter order

Part 1:

A signal has information bandwidth from 0 – 500Hz. It will be sampled with a 12-bit ADC.

The anti-aliasing filter is a Butterworth lowpass filter.

At 500Hz the signal must not be attenuated more than 3dB.

The anti-aliasing filter should attenuate the signal to a level equal to -72 dB at half the sampling frequency.

Calculate the minimum filter order required if the sampling frequency is 6kHz.

$$\text{Pass band: } G_p = -3\text{dB}$$

$$\omega_p = 3141.59\text{rad/s}$$

$$\text{Stop band: } \omega_s = \frac{\omega_{Fs}}{2} = 18849.6\text{rad/s}$$

$$G_s = -72\text{dB} \approx 12 \text{ bit} \times (-6\text{dB})$$

Problem 1 Determining the filter order

Calculate the minimum filter order and the resulting cut-off frequency in radians:

Pass band: $G_p = -3\text{dB}$

$$\omega_p = 3141.59\text{rad/s}$$

Stop band: $\omega_s = \frac{\omega_{Fs}}{2} = 18849.6\text{rad/s}$

$$G_s = -72\text{dB} \approx 12 \text{ bit} \times (-6\text{dB})$$

We would need a 5th order filter if we would like to attenuate by 72 dB at 3kHz.

The frequency normalized filter must be frequency scaled by a factor:

$$n = \frac{\log\left(\frac{10^{-\frac{G_s}{10}} - 1}{10^{-\frac{G_p}{10}} - 1}\right)}{2 \log\left(\frac{\omega_s}{\omega_p}\right)}$$

$$n = \frac{\log\left(\frac{10^{-\frac{-72}{10}} - 1}{10^{-\frac{-3}{10}} - 1}\right)}{2 \log\left(\frac{18849.6}{3141.59}\right)} = 4.63 \approx 5$$

$$\omega_c = \frac{\omega_p}{[10^{-G_p/10} - 1]^{1/2n}}$$

$$\omega_c = \frac{3141.59}{[10^{0.3} - 1]^{1/10}} = 3143.08 \text{ rad/s}$$

$$K_F = \frac{\omega_c}{1 \text{ rad/s}} = 3143.08$$

Problem 1 Determine the sampling frequency

Part 2:

We only have a 4th order Butterworth lowpass filter at our disposal.

Determine what sampling frequency F_s is required to obtain -72dB filter gain at $F_s/2$.

$$\frac{\omega_{FS}}{\omega_{3dB}} = 2^{\frac{N}{n}+1} = 2^{\frac{12}{4}+1} = 2^4 = 16$$

$$\omega_c = \frac{\omega_p}{[10^{-G_p/10} - 1]^{1/2n}}$$

$$\omega_c = \frac{3141.59}{[10^{-3/10} - 1]^{\frac{1}{2*4}}} = 3143.46 \text{ rad/s}$$

$$\omega_{FS} = 16 \times \omega_c = 16 \times 3143.46 \frac{\text{rad}}{\text{s}} = 50295.36 \frac{\text{rad}}{\text{s}}$$

$$F_s = \frac{\omega_{FS}}{2\pi} = 8 \text{ kHz}$$

Problem 1 Determine the attenuation obtained at the Nyquist frequency**Part 3:**

Conflicting design specifications defines a max sampling frequency of 4kHz.

What is the filter attenuation at $F_s/2$?

A 1st order filter has a high frequency slope of -6dB/octave.

A 4th order filter has a high frequency slope of -24dB/octave.

How many bits are toggled at this frequency?

-48dB equals 8 bits. There are 12 bits in the ADC,
hence the 4 LSB are toggled.

f [Hz]	$H _{dB}$
500	0dB
1000	-24dB
2000	-48dB
4000	-72dB

We see that we cannot avoid having some level of aliasing in the digitized data. The higher the sampling frequency the more insignificant the problem of aliasing will be.

We should always strive to reduce aliasing to an acceptable level. We do not want to sample an excessive amount of data, so we need to determine the smallest acceptable sampling frequency for our situation.

Trial and error: A typical approach is to perform a test sampling with a very high sampling frequency and then derive whatever quantitative features need to be measured on the over-sampled signal. Then a new trial is performed with the sampling frequency reduced by a factor of two. The same quantitative features are derived again. The procedure of reducing the sampling frequency is continued until one starts to observe unacceptable variance in the quantitative features measured on the digitized signals.

The minimum acceptable sampling frequency is then dependent on the nature of the quantitative features to be measured. Finding the time location of a peak or the time location of a maximum slope will require a higher sampling frequency than measuring the integral under the curve or the mean value.

Unused slides

To derive the rms value of the quantization noise we need to calculate the **mean and variance of a random variable X** .

Let X be a random variable with uniform probability distribution from a to b .

Then the expected mean value $\mu = E(X)$ is:

$$n_{rms} = \sqrt{\bar{n}^2 + \sigma_n^2}$$

$$\mu = E(X) = \frac{1}{b-a} \int_a^b X dX = \frac{1}{b-a} \left[\frac{X^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

The expected mean of the square $E(X^2)$ is:

$$\begin{aligned} E(X^2) &= \frac{1}{b-a} \int_a^b X^2 dX = \frac{1}{b-a} \left[\frac{X^3}{3} \right]_a^b = \frac{1}{3} \frac{1}{b-a} [b^3 - a^3] \\ &= \frac{1}{3} \frac{1}{b-a} (b-a)(b^2 + ab + a^2) = \frac{1}{3} (b^2 + ab + a^2) \end{aligned}$$

Variance of uniform distribution with zero mean.

The variance σ^2 is:

$$\sigma^2 \stackrel{\text{def}}{=} E(X^2) - \mu^2$$

$$\begin{aligned}\sigma^2 &\stackrel{\text{def}}{=} E(X^2) - \mu^2 = \frac{1}{3}(b^2 + ab + a^2) - \frac{(a+b)^2}{4} \\ &= \frac{b^3}{3} + \frac{ab}{3} + \frac{a^2}{3} - \frac{a^2}{4} - \frac{2ab}{4} - \frac{b^2}{4} \\ &= \frac{4b^3 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12} = \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}\end{aligned}$$

The rms value calculated using distributions:

$$E(X^2) = \mu^2 + \sigma^2$$

$$X_{rms} = \sqrt{E(X^2)} = \sqrt{\left(\frac{a+b}{2}\right)^2 + \frac{(b-a)^2}{12}} = \sqrt{\mu^2 + \sigma^2}$$

The rms value calculated using **time integrals**:

$$x_{an,rms} = \sqrt{\bar{x}_{an}^2 + \sigma_{x_{an}}^2}$$

Are these calculations related?

chatGPT:

I can calculate the rms value of a signal by calculating a time integral. But I can also calculate the rms value from a distribution of a random variable. How are they connected?

1. RMS from a Time Integral

For a **continuous-time signal** $x(t)$ over a period T , the RMS value is given by:

$$x_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

This formula treats the signal as a deterministic function of time.

2. RMS from a Probability Distribution

For a **random variable** X with a probability density function (PDF) $p(x)$, the RMS value is the **square root of the expected value of X^2** :

$$X_{\text{rms}} = \sqrt{\mathbb{E}[X^2]} = \sqrt{\int_{-\infty}^{\infty} x^2 p(x) dx}$$

This interpretation arises in stochastic processes, where X represents the amplitude of the signal at a random instant.

The Connection

The time-domain integral and the expectation integral are **formally the same** if we interpret the time integral as an expectation over time. If the signal $x(t)$ is a **stationary random process**, meaning its statistical properties don't change over time, then:

- The long-term time average of $x^2(t)$ converges to the statistical expectation of X^2 .
- The probability density function $p(x)$ describes how likely it is to observe a particular amplitude value.

In essence, when a signal is a **random process**, its RMS value can be computed using either method:

1. **Time-domain approach:** Taking a long enough observation period and averaging $x^2(t)$.
2. **Statistical approach:** Using the probability distribution of $x(t)$ and computing the expectation $\mathbb{E}[X^2]$.

Stationary random process: μ and σ^2 are time invariant.

We need to find out what effect averaging has on the mean and variance of a random variable.

Let us get help from AI.



Certainly! Let's derive the expressions for how averaging affects the **mean** and **variance** of a **uniformly distributed random variable**. We will consider a **uniformly distributed random variable** X with **mean** μ and **variance** σ^2 (which we denote as v here).

1. Definitions

- Let X be a **uniformly distributed random variable** with:
 - Mean: μ
 - Variance: v

2. Averaging n Samples

When we take n independent samples $X_1, X_2, \dots, X_n$ from the distribution of X , we can calculate the average of these samples:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

3. Mean of the Average

The **mean** of the average $\bar{X}$ can be found using the linearity of expectation:

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$$

Using the linearity of expectation:

$$E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \frac{n\mu}{n} = \mu$$

Conclusion for Mean

Thus, the **mean** of the average $\bar{X}$ remains the same as the **mean** of the original distribution:

$$\text{Mean of } \bar{X} = \mu$$

We see that if the random variable has a nonzero mean value, averaging will not change the mean value.

We also want to know what the effect of averaging is on the variance?

If we average M samples, the contribution to rms from the variance will be reduced, but not the contribution from the mean.

4. Variance of the Average

Next, we will derive the **variance** of the average $\bar{X}$. The **variance** of the average can be calculated as follows:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$$

Using the property that for independent random variables, the **variance** of their sum is the sum of their variances:

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n v = \frac{nv}{n^2} = \frac{v}{n}$$

Conclusion for Variance

Thus, the **variance** of the average $\bar{X}$ is reduced by a factor of n :

$$\text{Variance of } \bar{X} = \frac{v}{n}$$

$$X_{rms,M} = \sqrt{\mu^2 + \frac{\sigma^2}{M}}$$

Summary of Results

- **Mean of the Average:** The **mean** of the average $\bar{X}$ remains the same as the original **mean**:

$$\text{Mean of } \bar{X} = \mu$$

- **Variance of the Average:** The **variance** of the average $\bar{X}$ decreases inversely with the number of samples n :

$$\text{Variance of } \bar{X} = \frac{v}{n}$$

Implications

- **Averaging** tends to keep the expected value (**mean**) unchanged while reducing the variability (**variance**) of the estimates. This makes averaging an effective technique for improving the reliability of measurements derived from random variables.

Consequence of inadequate amplification

We saw on the previous slide that an inadequate gain will reduce the utilization of the ADC voltage range and therefore also the utilization of some bits. We can express this mathematically:

If we only use 50% of max gain,
we lose the use of 1bit.

$$\alpha = 1/2$$

$$\alpha = 2^{-K} \Rightarrow -K \log(2) = \log(\alpha)$$

$$K = \frac{-\log(\alpha)}{\log(2)} = \frac{\log(1/\alpha)}{\log(2)}$$

If we only use 10% of max gain,
we lose the use of 3.3bit.

$$\alpha = 0.1$$

$$K = \frac{\log(2)}{\log(2)} = 1 \text{ bit}$$

$$K = \frac{\log(10)}{\log(2)} \approx 3.3 \text{ bit}$$

The fewer bits you have in your ADC, the more important it is to provide adequate amplification. On the other hand, too high amplification will cause the amplified signal to exceed the voltage range of the ADC. This will cause **clipping**, as the digital level cannot exceed the full range.

The effects of clipping are serious and renders the digitized data useless. If the maximum amplitude of a signal is hard to predict, it is necessary to allow some **head room** to avoid clipping. If we used a 16-bit ADC instead of a 10-bit ADC, we could use the 6 MSB as head room, while maintaining the same amplitude resolution.

The effective number of bits (ENOB)

In an ideal ADC the smallest amplitude that can be converted is:

$$\Delta V_{LSB} = \frac{V_{range}}{2^N}$$

In decibel the amplitude range covered by an ADC is called its **dynamic range**:

$$DR_{theory} = 6.02N \text{ dB}$$

In a real ADC, the DR is slightly less than the ideal value. Hence one can calculate the **effective number of bits** (ENOB) that would correspond to the real-life DR.

$$ENOB = \frac{DR_{real}}{6.02}$$

$$SNR = 6.02N + 20 \log(\alpha) + 20 \log \sqrt{M} - 4.26 \text{ dB} + 10.79 \text{ dB}$$

$$SNR = DR_{real} + 20 \log(\alpha) + 20 \log \sqrt{M} - 4.26 \text{ dB} + 10.79 \text{ dB}$$

Example:

A non-ideal 10-bit truncating ADC is said in the data sheet to have a DR = 55 dB. Assuming the full voltage scale is used, and no oversampling is done, what is the effective number of bits?

$$ENOB = \frac{DR_{real}}{6.02} = \frac{55}{6.02} = 9.14$$

$$SNR = 55 + 20 \log(1) + 20 \log \sqrt{1} - 4.26 \text{ dB} + 10.79 \text{ dB}$$

$$= 61.5 \text{ dB}$$

A reduction from 66.73 dB

What is the SNR?